Table S3

Table S3. List of biophysical profiles and simulation results for the examination of KJE and GroELS system sul

Parameters						*No chaperones*	
k_m	K_m	k_f	K_f	k_a	C_{crit}	Native protein	Aggregated protein
1.420007	-1.22838	0.59514	2.470275	-2.37376	-2.7363	86.21742429	1.10E-08
-1.04836	0.96031	2.003869	2.561892	2.762631	0.280836	59.82033857	0.003486338
0.049801	-1.8113	1.952618	7.534692	2.826289	2.73751	199.4562706	1.36E-10
-0.11592	2.629487	-1.37632	6.655974	0.477679	2.081409	9.593492538	108.0061693
0.446656	-1.61119	2.125	5.229832	2.988615	1.042851	198.9160397	2.30E-10
1.936378	1.403042	2.873412	3.011459	0.617478	0.507457	17.96739247	13.51269567
-2.77691	2.048923	-1.35605	2.405496	2.992081	0.735717	57.80288545	1.41071E-05
-0.56364	-1.29839	-1.71815	3.21044	-2.20203	2.230603	50.07641354	0.000766961
-1.67378	-2.21229	-0.64312	3.277437	-2.18097	2.19536	128.2459132	1.55E-06
1.576094	0.360729	2.819752	4.485247	-2.13991	1.084404	189.3971424	4.89601E-05
1.202419	2.602818	2.311731	5.909439	0.842358	-0.76015	177.3171578	4.71E-06
-1.72786	1.602557	2.251704	2.935977	-1.7189	2.03496	124.6034087	0.004825796
-2.29495	-0.11479	0.035125	6.490567	-0.05642	0.695783	182.6858386	7.21E-08
2.901537	-1.61097	0.723297	7.482676	-0.35088	-0.78811	195.8749411	0
-1.61485	0.599904	1.141919	4.394615	0.531164	1.424325	193.1451994	2.43E-06
-2.64511	1.353848	-0.66003	5.13212	-1.21612	-0.60383	143.4465041	1.74E-08
1.340056	2.72505	-0.6881	5.985382	2.344147	-2.33754	3.928926522	3.14435E-05
-1.95063	0.430992	-1.16548	6.147013	1.59649	1.651427	94.01625068	0.006615322
-0.48746	2.809899	0.492445	7.065221	1.367461	0.972385	176.0114904	7.55621E-05
-0.15875	1.171797	2.341707	5.408391	2.451547	-2.19704	196.9444234	2.09E-09
0.382279	-1.36161	1.784874	5.981799	0.636948	2.767183	199.2308235	6.31E-09
1.634931	-0.36976	-0.26129	3.470243	-2.87082	-2.28199	140.8039773	9.41E-08
1.666085	1.038777	-1.49165	6.848615	-0.26273	-1.3401	13.06084863	0.000197279
-0.16038	-2.32529	-2.20651	4.903593	2.897445	2.140365	25.10446321	1.10959E-05
-2.96659	-0.73476	1.881436	7.756389	-1.39526	-1.78349	199.4177823	0
-1.57666	-0.6651	0.047628	6.790063	2.947213	-2.37579	182.1375151	3.39E-10
-2.9614	2.812558	0.532391	3.341018	-0.07815	1.594617	161.1365263	8.73E-07
-2.50024	-1.91383	-0.45471	2.330445	1.92023	0.94758	71.99237247	2.29E-06
-1.84861	-0.22492	0.998388	6.120202	-1.12718	1.665564	197.3771365	5.59E-08
1.911181	-2.26951	-2.21438	4.889813	-1.99622	-2.88928	24.75504359	1.34E-10
-0.01009	-0.97289	0.967689	6.974398	-2.25123	-0.07673	197.3066028	8.59E-10
2.651158	0.974257	-0.20095	6.608851	1.699514	0.144424	95.02498094	0.00058416
1.02409	-1.11348	-2.19356	3.943946	-0.41302	-2.16705	24.31366337	1.28E-07
1.882108	-2.7246	-2.84937	3.145124	-1.06385	-0.87217	7.312707847	2.93E-09
-1.27106	1.750073	0.647246	7.182784	-2.81383	0.17389	193.0452468	1.68E-07
-1.04873	2.231478	-2.06203	4.497723	2.774708	-1.59317	12.70494619	6.73988E-05
-1.29433	-2.88203	0.432699	4.159328	-0.14104	-2.21293	186.5641752	0
2.929863	1.028485	-0.48226	5.065976	-1.87415	-2.70589	65.74421752	1.95E-06
2.223318	0.685276	2.997893	6.037349	1.503883	-2.62137	199.0224333	0
-2.44291	-0.32196	-2.02678	4.060396	0.509297	2.716252	31.73947515	3.135590159
0.502266	2.930275	1.478806	6.906772	2.518008	0.911147	180.4387918	6.76264E-05
2.503519	2.857471	2.508216	7.671116	-2.53327	1.24337	169.8178213	0.000403465
0.286536	1.795356	-0.04871	4.959227	-2.24352	-0.97582	77.59101629	8.78698E-05

Table S5

Table S5. List of biophysical profiles and simulation results for the examination of KJE and GroELS system sub

Parameters						*No chaperones*	
k_m	K_m	k_f	K_f	k_a	C_{crit}	Native protein	Aggregated protein
1.420007	-1.22838	0.59514	2.470275	-2.37376	-2.7363	1721.966243	4.42E-06
-1.04836	0.96031	2.003869	2.561892	2.762631	0.280836	698.2210701	1036.602426
0.049801	-1.8113	1.952618	7.534692	2.826289	2.73751	1994.859603	1.36E-08
-0.11592	2.629487	-1.37632	6.655974	0.477679	2.081409	98.87022092	1721.042122
0.446656	-1.61119	2.125	5.229832	2.988615	1.042851	1989.643007	2.30E-08
1.936378	1.403042	2.873412	3.011459	0.617478	0.507457	44.94326389	1712.374346
-2.77691	2.048923	-1.35605	2.405496	2.992081	0.735717	1587.05553	102.622792
-0.56364	-1.29839	-1.71815	3.21044	-2.20203	2.230603	1630.590377	55.39246481
-1.67378	-2.21229	-0.64312	3.277437	-2.18097	2.19536	1748.891132	0.000335394
1.576094	0.360729	2.819752	4.485247	-2.13991	1.084404	1921.727469	0.005481124
1.202419	2.602818	2.311731	5.909439	0.842358	-0.76015	1829.531857	0.00178622
-1.72786	1.602557	2.251704	2.935977	-1.7189	2.03496	1639.30928	120.8360407
-2.29495	-0.11479	0.035125	6.490567	-0.05642	0.695783	1880.327376	1.69906E-05
2.901537	-1.61097	0.723297	7.482676	-0.35088	-0.78811	1962.848569	2.71E-09
-1.61485	0.599904	1.141919	4.394615	0.531164	1.424325	1941.645537	0.000463638
-2.64511	1.353848	-0.66003	5.13212	-1.21612	-0.60383	1767.594443	2.84951E-05
1.340056	2.72505	-0.6881	5.985382	2.344147	-2.33754	812.497764	636.2475742
-1.95063	0.430992	-1.16548	6.147013	1.59649	1.651427	1492.790538	229.7057457
-0.48746	2.809899	0.492445	7.065221	1.367461	0.972385	1780.039087	76.60893185
-0.15875	1.171797	2.341707	5.408391	2.451547	-2.19704	1970.875138	2.55E-07
0.382279	-1.36161	1.784874	5.981799	0.636948	2.767183	1992.653646	6.33E-07
1.634931	-0.36976	-0.26129	3.470243	-2.87082	-2.28199	1768.57928	1.60836E-05
1.666085	1.038777	-1.49165	6.848615	-0.26273	-1.3401	723.0356755	937.9347774
-0.16038	-2.32529	-2.20651	4.903593	2.897445	2.140365	503.448707	1185.874957
-2.96659	-0.73476	1.881436	7.756389	-1.39526	-1.78349	1994.486346	0
-1.57666	-0.6651	0.047628	6.790063	2.947213	-2.37579	1877.960231	5.05E-08
-2.9614	2.812558	0.532391	3.341018	-0.07815	1.594617	1818.277518	0.002270945
-2.50024	-1.91383	-0.45471	2.330445	1.92023	0.94758	1704.404227	0.002763057
-1.84861	-0.22492	0.998388	6.120202	-1.12718	1.665564	1975.410201	6.27E-06
1.911181	-2.26951	-2.21438	4.889813	-1.99622	-2.88928	1660.555108	5.86E-07
-0.01009	-0.97289	0.967689	6.974398	-2.25123	-0.07673	1974.902653	8.62E-08
2.651158	0.974257	-0.20095	6.608851	1.699514	0.144424	651.6862062	1104.613652
1.02409	-1.11348	-2.19356	3.943946	-0.41302	-2.16705	1659.68895	0.000582953
1.882108	-2.7246	-2.84937	3.145124	-1.06385	-0.87217	1563.77961	0.000141346
-1.27106	1.750073	0.647246	7.182784	-2.81383	0.17389	1938.185047	2.50809E-05
-1.04873	2.231478	-2.06203	4.497723	2.774708	-1.59317	194.4471809	1445.335823
-1.29433	-2.88203	0.432699	4.159328	-0.14104	-2.21293	1902.505614	0
2.929863	1.028485	-0.48226	5.065976	-1.87415	-2.70589	1695.270371	0.001291295
2.223318	0.685276	2.997893	6.037349	1.503883	-2.62137	1990.656853	4.58E-09
-2.44291	-0.32196	-2.02678	4.060396	0.509297	2.716252	1217.121675	462.4745592
0.502266	2.930275	1.478806	6.906772	2.518008	0.911147	1804.729003	87.50219933
2.503519	2.857471	2.508216	7.671116	-2.53327	1.24337	1825.515193	0.200825459
0.286536	1.795356	-0.04871	4.959227	-2.24352	-0.97582	1673.922863	8.230653767

Supplemental Information for "FoldEco: A Model for Proteostasis in *E. coli*"

Evan T. Powers[1], David L. Powers[2], and Lila M. Gierasch[3]

[1]Department of Chemistry, The Scripps Research Institute, La Jolla, CA 92037

[2]Department of Mathematics and Computer Science, Clarkson University, Potsdam, NY 13699

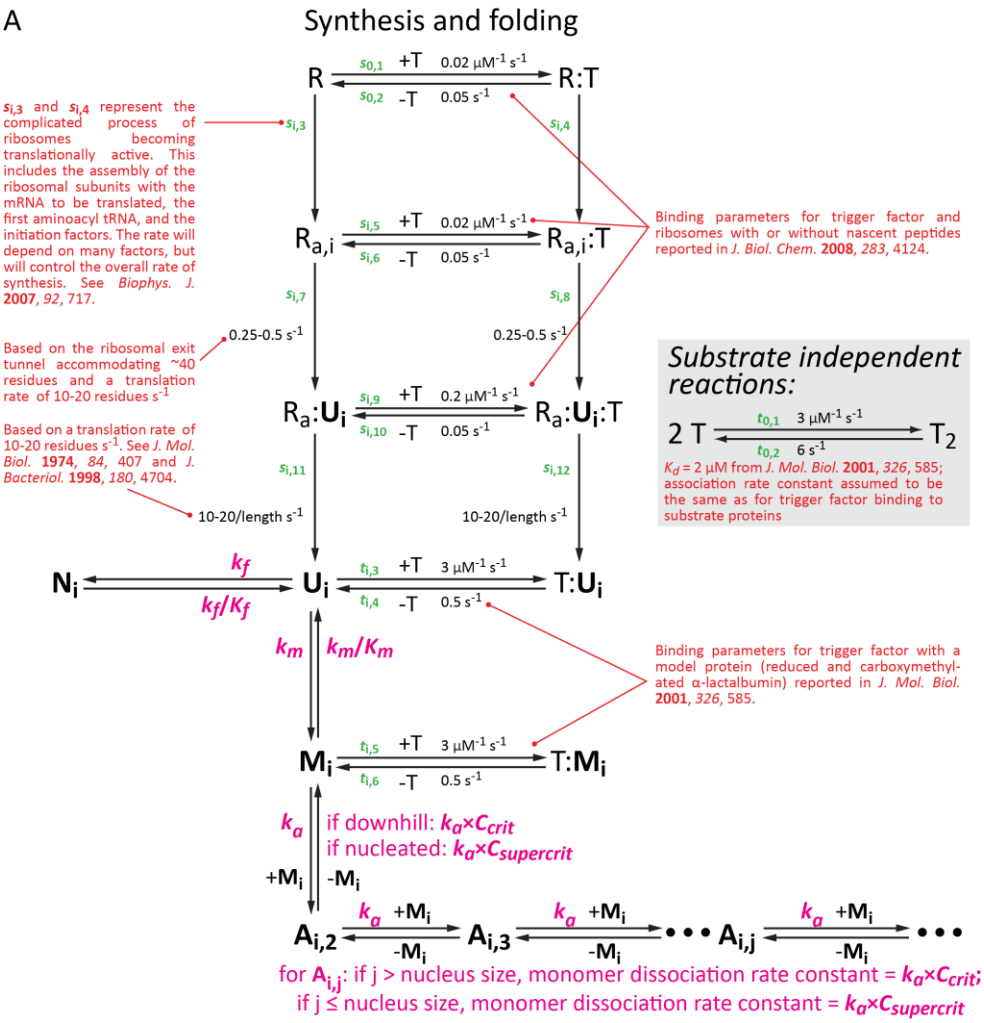
[3]Departments of Chemistry and Biochemistry & Molecular Biology, University of Massachusetts-Amherst, Amherst MA 01003

Correspondence to: epowers@scripps.edu; gierasch@biochem.umass.edu

Supplemental Figure Legends

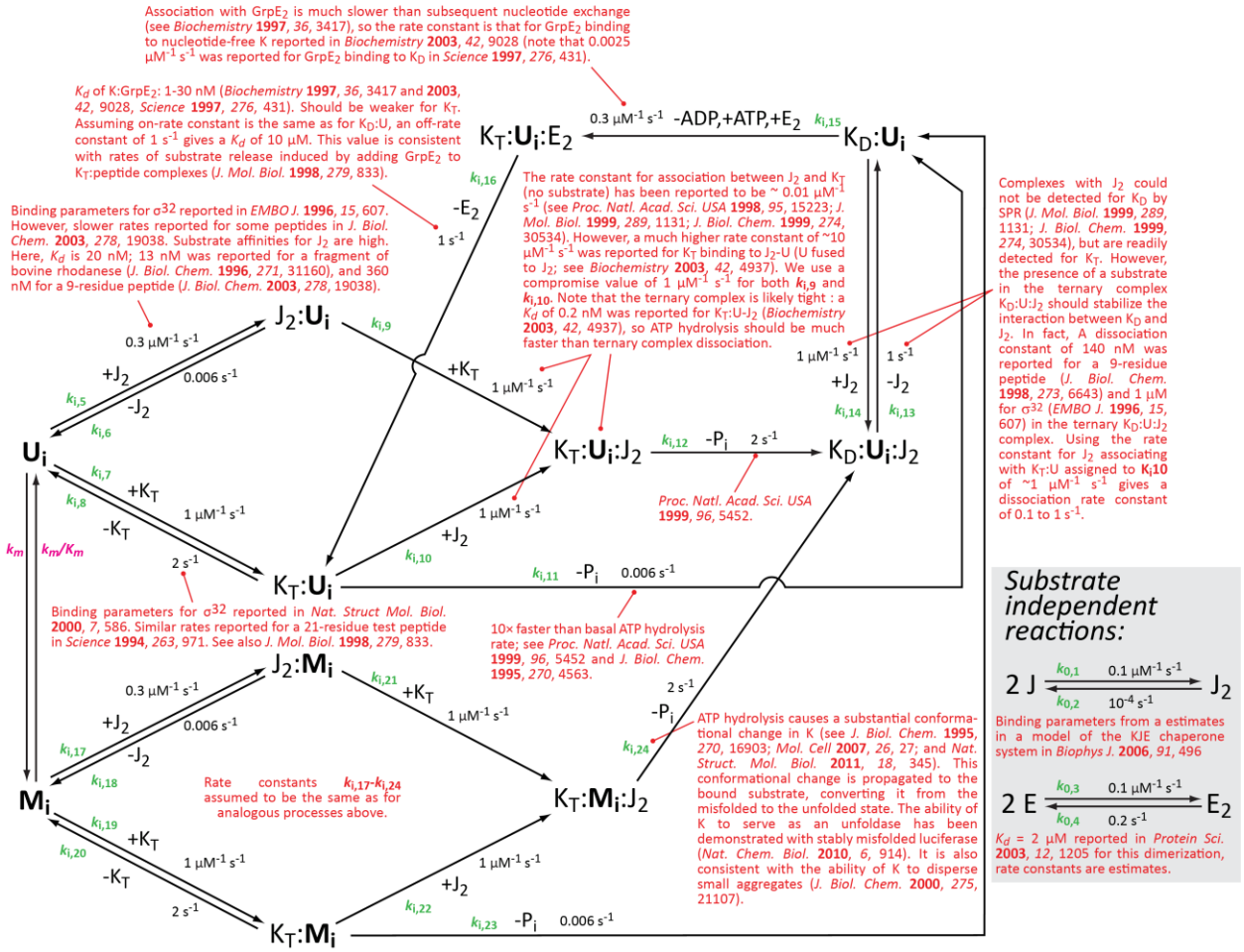
Figure S1, related to Figure 1. The FoldEco model for protein folding in *E. coli* in detail. The variable names for the various rate constants are in green text. Estimated values for each rate constant based on literature data for each step are shown next to the arrows. Explanations for the choice of rate constants and other information are in red text. Complexes between proteostasis network components and the client protein are denoted by abbreviations for the species in the complex separated by colons. (A) Synthesis and folding system in FoldEco. Key: R = ribosome; $R_{a,i}$ = ribosome actively synthesizing client protein i; R_a:U_i = ribosome with emerging nascent chain; T, T_2 = trigger factor monomer and dimer; U_i = unfolded client protein; M_i = misfolded client protein; N_i = native protein; $A_{i,j}$ = aggregates consisting of j monomers. (B) KJE system. K_T = ATP-bound DnaK; K_D = ADP-bound DnaK; J, J_2 = DnaJ monomer and dimer; E, E_2 = GrpE monomer and dimer. (C) GroELS system. GrL, GrL_T, GrL_D = nucleotide free, ATP-bound, and ADP bound GroEL; GrS = GroES. When the states of the *cis* and *trans* cavities of GroEL are both defined, they are separated by two colons. (D) B+KJE system. B = ClpB. (E) Degradation system. Lon = Lon protease for unfolded and misfolded proteins; Dn = ClpAP-type protease for natively folded proteins with degrons.

A


Synthesis and folding
R
R:T
$s_{0,1}$ +T 0.02 μM^{-1} s^{-1}
$s_{0,2}$ -T 0.05 s^{-1}
$s_{i,3}$ and $s_{i,4}$ represent the complicated process of ribosomes becoming translationally active. This includes the assembly of the ribosomal subunits with the mRNA to be translated, the first aminoacyl tRNA, and the initiation factors. The rate will depend on many factors, but will control the overall rate of synthesis. See Biophys. J. 2007, 92, 717.
$s_{i,3}$
$s_{i,4}$
$R_{a,i}$
$R_{a,i}$:T
$s_{i,5}$ +T 0.02 μM^{-1} s^{-1}
$s_{i,6}$ -T 0.05 s^{-1}
Binding parameters for trigger factor and ribosomes with or without nascent peptides reported in J. Biol. Chem. 2008, 283, 4124.
$s_{i,7}$
$s_{i,8}$
Based on the ribosomal exit tunnel accommodating ~40 residues and a translation rate of 10-20 residues s^{-1}
0.25-0.5 s^{-1}
0.25-0.5 s^{-1}
Substrate independent reactions:
2 T $t_{0,1}$ 3 μM^{-1} s^{-1} / $t_{0,2}$ 6 s^{-1} T_2
K_d = 2 μM from J. Mol. Biol. 2001, 326, 585; association rate constant assumed to be the same as for trigger factor binding to substrate proteins
R_a:U_i
R_a:U_i:T
$s_{i,9}$ +T 0.2 μM^{-1} s^{-1}
$s_{i,10}$ -T 0.05 s^{-1}
Based on a translation rate of 10-20 residues s^{-1}. See J. Mol. Biol. 1974, 84, 407 and J. Bacteriol. 1998, 180, 4704.
$s_{i,11}$
$s_{i,12}$
10-20/length s^{-1}
10-20/length s^{-1}
k_f
k_f/K_f
N_i
U_i
T:U_i
$t_{i,3}$ +T 3 μM^{-1} s^{-1}
$t_{i,4}$ -T 0.5 s^{-1}
k_m
k_m/K_m
Binding parameters for trigger factor with a model protein (reduced and carboxymethylated α-lactalbumin) reported in J. Mol. Biol. 2001, 326, 585.
M_i
T:M_i
$t_{i,5}$ +T 3 μM^{-1} s^{-1}
$t_{i,6}$ -T 0.5 s^{-1}
k_a
if downhill: $k_a \times C_{crit}$
if nucleated: $k_a \times C_{supercrit}$
+M_i
-M_i
$A_{i,2}$
$A_{i,3}$
$A_{i,j}$
k_a +M_i
-M_i
for $A_{i,j}$: if j > nucleus size, monomer dissociation rate constant = $k_a \times C_{crit}$;
if j ≤ nucleus size, monomer dissociation rate constant = $k_a \times C_{supercrit}$

B

DnaK/DnaJ/GrpE (KJE) system



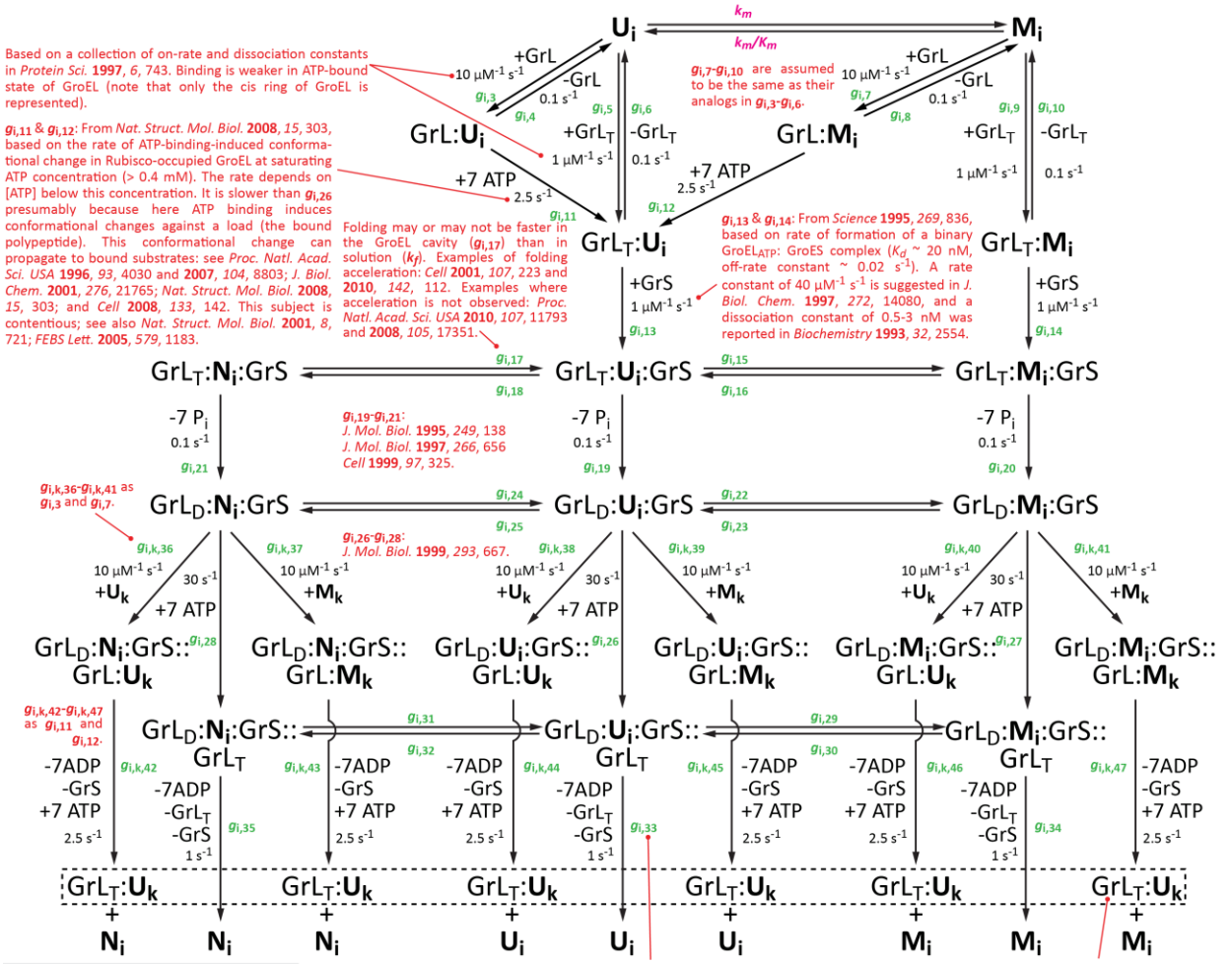
Substrate independent reactions:

Binding parameters from a estimates in a model of the KJE chaperone system in *Biophys J.* **2006**, *91*, 496

K_d = 2 μM reported in *Protein Sci.* **2003**, *12*, 1205 for this dimerization, rate constants are estimates.

C

GroEL/GroES (GroELS) system



Based on a collection of on-rate and dissociation constants in *Protein Sci.* **1997**, *6*, 743. Binding is weaker in ATP-bound state of GroEL (note that only the cis ring of GroEL is represented).

$g_{i,11}$ & $g_{i,12}$: From *Nat. Struct. Mol. Biol.* **2008**, *15*, 303, based on the rate of ATP-binding-induced conformational change in Rubisco-occupied GroEL at saturating ATP concentration (> 0.4 mM). The rate depends on [ATP] below this concentration. It is slower than $g_{i,26}$ presumably because here ATP binding induces conformational changes against a load (the bound polypeptide). This conformational change can propagate to bound substrates: see *Proc. Natl. Acad. Sci. USA* **1996**, *93*, 4030 and **2007**, *104*, 8803; *J. Biol. Chem.* **2001**, *276*, 21765; *Nat. Struct. Mol. Biol.* **2008**, *15*, 303; and *Cell* **2008**, *133*, 142. This subject is contentious; see also *Nat. Struct. Mol. Biol.* **2001**, *8*, 721; *FEBS Lett.* **2005**, *579*, 1183.

$g_{i,7}$-$g_{i,10}$ are assumed to be the same as their analogs in $g_{i,3}$-$g_{i,6}$.

Folding may or may not be faster in the GroEL cavity ($g_{i,17}$) than in solution (k_f). Examples of folding acceleration: *Cell* **2001**, *107*, 223 and **2010**, *142*, 112. Examples where acceleration is not observed: *Proc. Natl. Acad. Sci. USA* **2010**, *107*, 11793 and **2008**, *105*, 17351.

$g_{i,13}$ & $g_{i,14}$: From *Science* **1995**, *269*, 836, based on rate of formation of a binary $GroEL_{ATP}$: GroES complex (K_d ~ 20 nM, off-rate constant ~ 0.02 s^{-1}). A rate constant of 40 $\mu M^{-1}\ s^{-1}$ is suggested in *J. Biol. Chem.* **1997**, *272*, 14080, and a dissociation constant of 0.5-3 nM was reported in *Biochemistry* **1993**, *32*, 2554.

$g_{i,19}$-$g_{i,21}$: *J. Mol. Biol.* **1995**, *249*, 138 *J. Mol. Biol.* **1997**, *266*, 656 *Cell* **1999**, *97*, 325.

$g_{i,k,36}$-$g_{i,k,41}$ as $g_{i,3}$ and $g_{i,7}$.

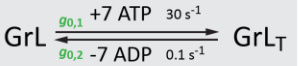
$g_{i,26}$-$g_{i,28}$: *J. Mol. Biol.* **1999**, *293*, 667.

$g_{i,k,42}$-$g_{i,k,47}$ as $g_{i,11}$ and $g_{i,12}$.

$g_{i,33}$-$g_{i,35}$: Rate of substrate ejection after ATP binding to trans ring, uncatalyzed by peptide binding. *J. Mol. Biol.* **1995**, *249*, 138, *Cell* **1999**, *97*, 325. See also *FEBS Lett.* **2010**, *584*, 951.

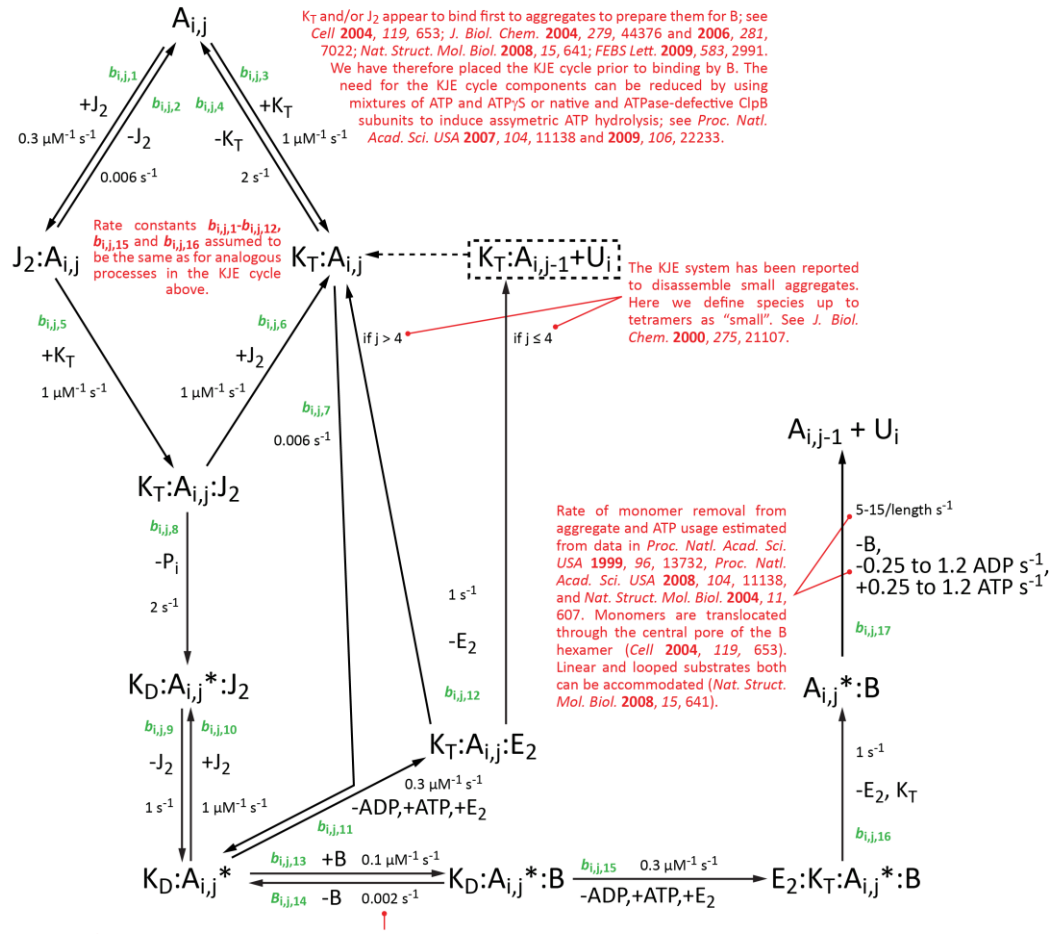
Switch index i to k (if k ≠ i), re-renter cycle at $g_{i,13}$.

Substrate independent reactions:



$g_{0,1}$ & $g_{0,2}$ are not on- and off-rates for ATP and ADP, but entry and exit rates from unproductive (i.e., without substrate) GroELS cycles; "entry-rate" corresponds to the slow phase (k_1) from *J. Mol. Biol.* **1999**, *293*, 667. "Exit-rate" assumes that ATP hydrolysis is rate limiting (compared to GroES binding, etc.).

D ClpB/DnaK/DnaJ/GrpE (B+KJE) disaggregation system



K_T and/or J_2 appear to bind first to aggregates to prepare them for B; see *Cell* **2004**, *119*, 653; *J. Biol. Chem.* **2004**, *279*, 44376 and **2006**, *281*, 7022; *Nat. Struct. Mol. Biol.* **2008**, *15*, 641; *FEBS Lett.* **2009**, *583*, 2991. We have therefore placed the KJE cycle prior to binding by B. The need for the KJE cycle components can be reduced by using mixtures of ATP and ATPγS or native and ATPase-defective ClpB subunits to induce assymetric ATP hydrolysis; see *Proc. Natl. Acad. Sci. USA* **2007**, *104*, 11138 and **2009**, *106*, 22233.

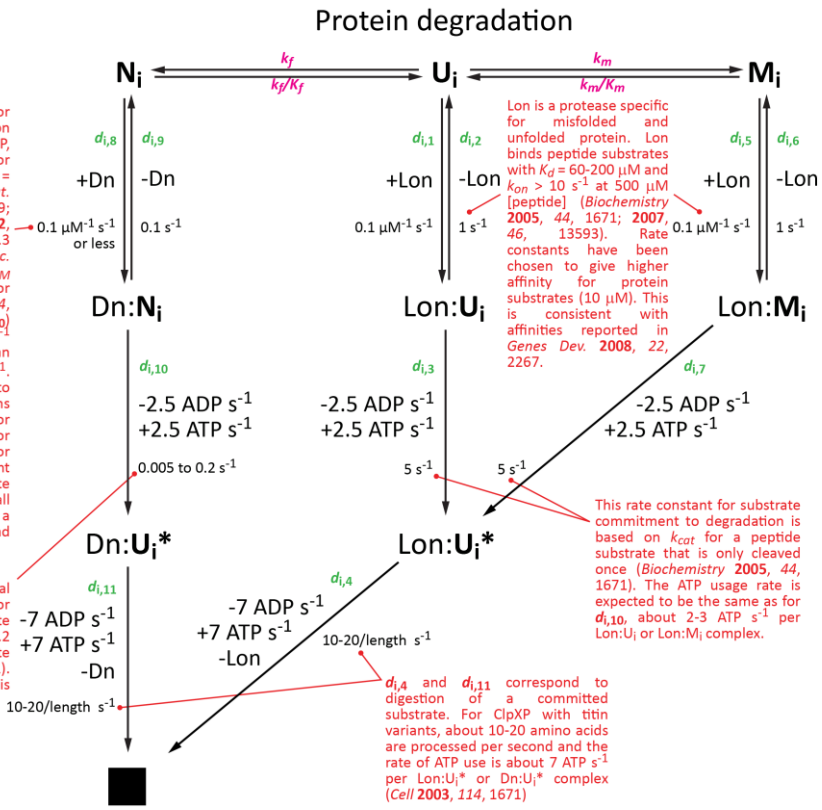
Rate constants $b_{i,j,1}$-$b_{i,j,12}$, $b_{i,j,15}$ and $b_{i,j,16}$ assumed to be the same as for analogous processes in the KJE cycle above.

The KJE system has been reported to disassemble small aggregates. Here we define species up to tetramers as "small". See *J. Biol. Chem.* **2000**, *275*, 21107.

Rate of monomer removal from aggregate and ATP usage estimated from data in *Proc. Natl. Acad. Sci. USA* **1999**, *96*, 13732, *Proc. Natl. Acad. Sci. USA* **2008**, *104*, 11138, and *Nat. Struct. Mol. Biol.* **2004**, *11*, 607. Monomers are translocated through the central pore of the B hexamer (*Cell* **2004**, *119*, 653). Linear and looped substrates both can be accommodated (*Nat. Struct. Mol. Biol.* **2008**, *15*, 641).

K_d for ClpB w/ peptide substrates reported as ~20 nM in *Nat. Struct. Mol. Biol.* **2004**, *11*, 607. The association rate constant is assumed to be similar to that for ClpXP and ClpAP binding to substrates (see Degradation). Much weaker affinity (10 μM) was reported in *FEBS Lett.* **2009**, *583*, 2991. The "hand-off" from K to B appears to be mediated by the M domain on B (*Mol. Cell* **2007**, *25*, 247). One peptide binds per ClpB hexamer (*Nat. Struct. Mol. Biol.* **2004**, *11*, 607).

E



"Dn" is a protease specific for folded proteins with degradation signals (degrons), e.g., ClpXP, ClpAP, HslUV. Typical K_M values for ClpAP, ClpXP and HslUV are K_M = 1-10 μM (*Cell* **2003**, *114*, 511; *Nat. Struct. Mol. Biol.* **2008**, *15*, 139; *Nat. Struct. Mol. Biol.* **2005**, **12**, 245). Off-rate constants are < 0.3 s^{-1} for ClpA (*J. Am. Chem. Soc.* **2007**, *129*, 12379) and K_d and K_M are reported to be comparable for these proteases (*Cell* **2003**, *114*, 511), implying that k_{cat} (i.e., $d_{i,10}$) << k_{off} (i.e., $d_{i,9}$). So, taking 0.1 s^{-1} for the off-rate constant gives an on-rate constant of 0.1 μM^{-1} s^{-1}. This rate constant applies only to proteins with pre-existing degrons that interact directly with Dn. For proteins without a degron or proteins that require an adaptor (e.g., ClpS) the rate constant should be equal to that of the rate limiting step in the overall degradation process (installing a degron, adaptor binding, and delivery to protease).

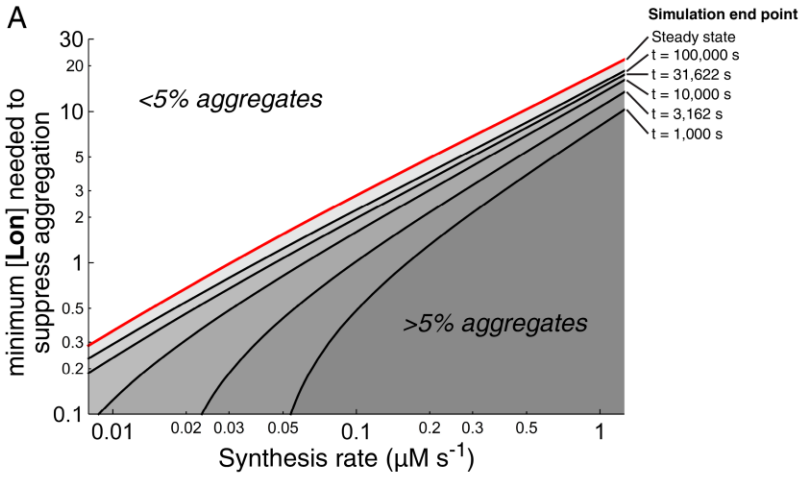
$d_{i,10}$ corresponds to mechanical unfolding of a native substrate. For ClpXP with titin variants, the rate constant ranges from 0.005 to 0.2 s^{-1}, depending on substrate stability (*Cell* **2003**, *114*, 1671). The specific rate of ATP usage is 2-3 ATP s^{-1} per Dn:N_i complex.

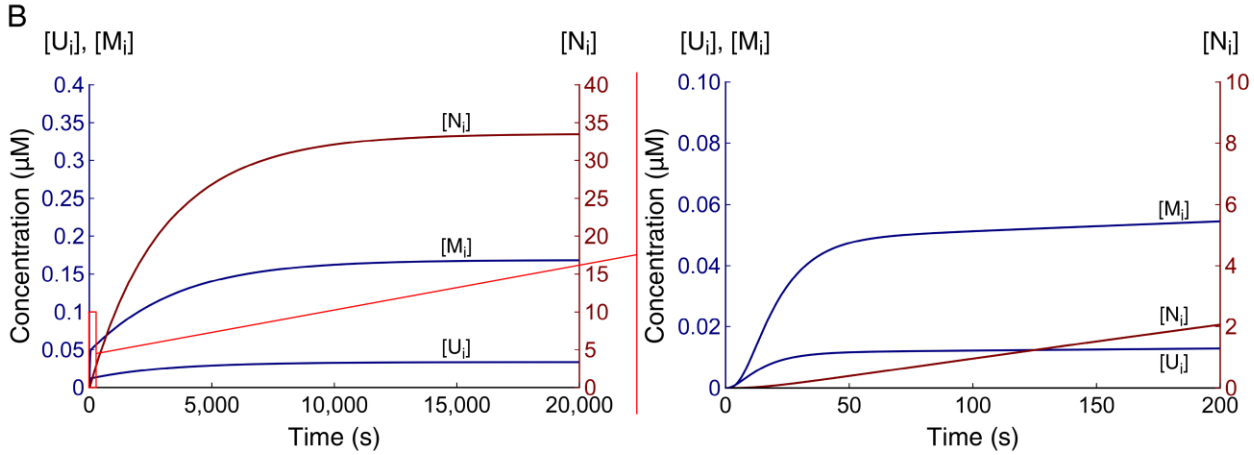
Lon is a protease specific for misfolded and unfolded protein. Lon binds peptide substrates with K_d = 60-200 μM and k_{on} > 10 s^{-1} at 500 μM [peptide] (*Biochemistry* **2005**, *44*, 1671; **2007**, *46*, 13593). Rate constants have been chosen to give higher affinity for protein substrates (10 μM). This is consistent with affinities reported in *Genes Dev.* **2008**, *22*, 2267.

This rate constant for substrate commitment to degradation is based on k_{cat} for a peptide substrate that is only cleaved once (*Biochemistry* **2005**, *44*, 1671). The ATP usage rate is expected to be the same as for $d_{i,10}$, about 2-3 ATP s^{-1} per Lon:U_i or Lon:M_i complex.

$d_{i,4}$ and $d_{i,11}$ correspond to digestion of a committed substrate. For ClpXP with titin variants, about 10-20 amino acids are processed per second and the rate of ATP use is about 7 ATP s^{-1} per Lon:U_i* or Dn:U_i* complex (*Cell* **2003**, *114*, 1671)

Figure S2, related to Figure 2. Kinetics of protein accumulation in FoldEco. (A) As Figure 2, except the black curves represent the minimum concentration of Lon needed to suppress aggregation to <5% of total protein after 1,000 s, 3,162 s, 10,000 s, 31,622 s, or 100,000 s, as indicated. The red curve represents the minimum concentration of Lon required to suppress aggregation altogether at steady state (which is achieved by keeping the steady state concentration of M_i below its critical concentration of 0.1 μM). (B) Concentrations of U_i, M_i (blue curves, left-hand axis) and N_i (red curve, right-hand axis) vs time. See Extended Experimental Procedures for parameters used in the simulation. (C) Simplified model for analyzing protein accumulation kinetics. Lowercase greek letters represent rate constants. Synthesis is condensed to a single step with rate σ. Degradation is simplified to a Briggs-Haldane scheme, where D is a Lon-type protease that degrades U and M. The rate constants for substrate binding, release, and degradation (δ_1, δ_2, δ_3) are assumed to be the same for U and M. The folding and unfolding rate constants are φ_1 and φ_2, while those for misfolding and un-misfolding are μ_1 and μ_2.


A
Simulation end point
Steady state
t = 100,000 s
t = 31,622 s
t = 10,000 s
t = 3,162 s
t = 1,000 s
<5% aggregates
>5% aggregates
minimum [Lon] needed to suppress aggregation
Synthesis rate (μM s-1)


B
[Ui], [Mi]
[Ni]
Concentration (μM)
Time (s)

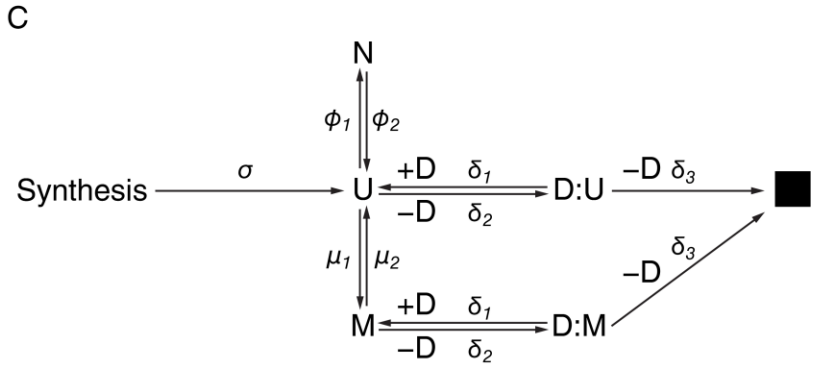

C
N
Synthesis
σ
U
D:U
M
D:M
+D
−D
ϕ1
ϕ2
μ1
μ2
δ1
δ2
δ3

Figure S3, related to Figure 3. Biophysical profiles for various groups of client proteins from the simulation at a low synthesis rate. (A) As Figure 3A, but for proteins that aggregate at a low synthesis rate in the absence of chaperones. (B) As Figure 3A, but for the top KJE system substrates in the presence of the GroELS system. (C) As Figure 3A, but for the top GroELS system substrates in the presence of the KJE system. (D) As Figure 3A, but for the client proteins corresponding to the blue data points in region 1 of Figure 3C, for which the combined effects of the KJE and GroELS systems are less than the summed effects, but neither system alone completely rescues the client protein. (E) As Figure 3A, but for the client proteins corresponding to the blue data points in region 2 of Figure 3C, for which the combined effects of the KJE and GroELS systems are similar.

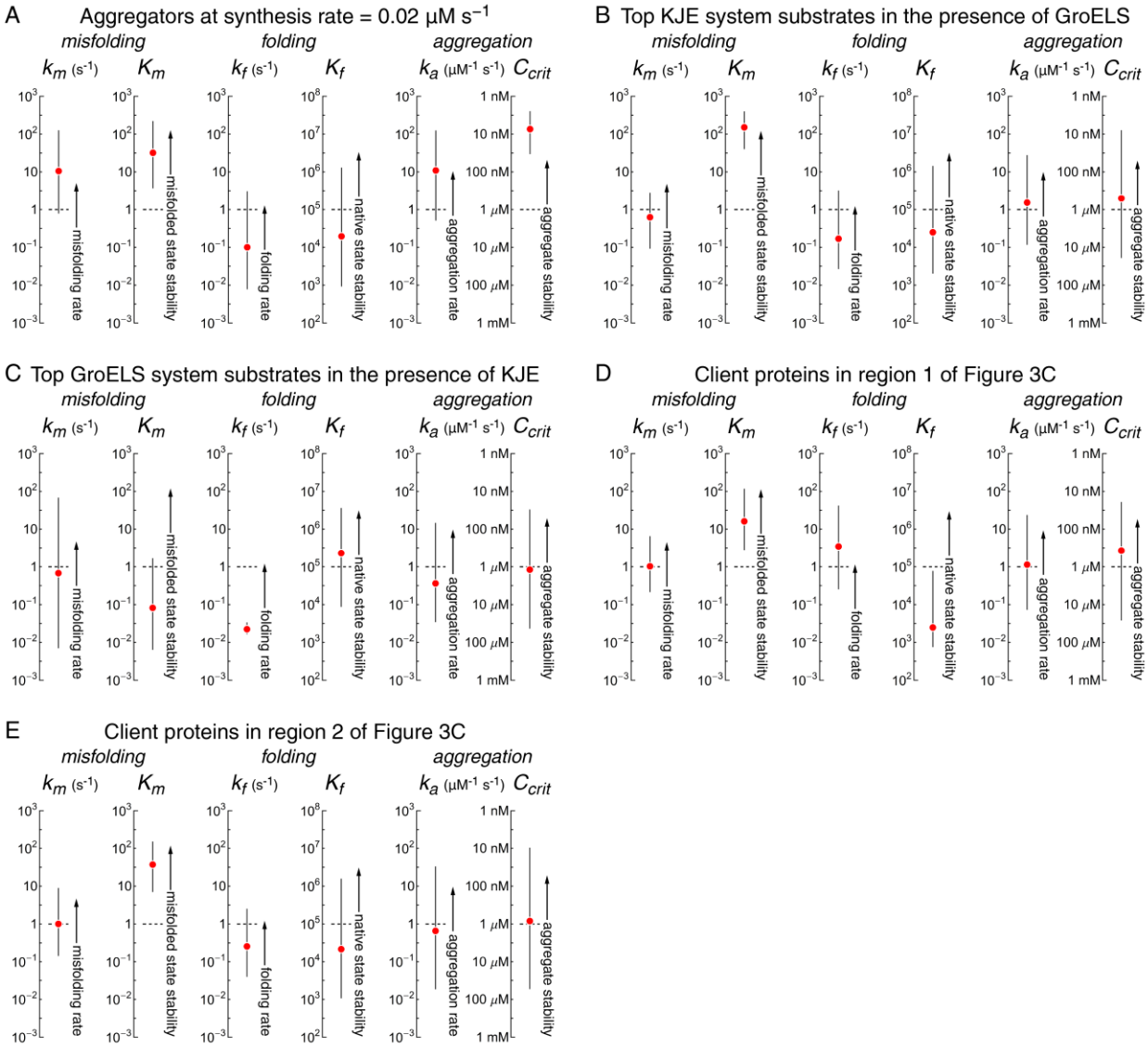


Figure S4, related to Figure 4. Biophysical profiles for various groups of client proteins from the simulation at a high synthesis rate. (A) As Figure 4A, but for the top KJE system substrates in the presence of the GroELS system. (B) As Figure 4B, but for the top GroELS system substrates in the presence of the KJE system.

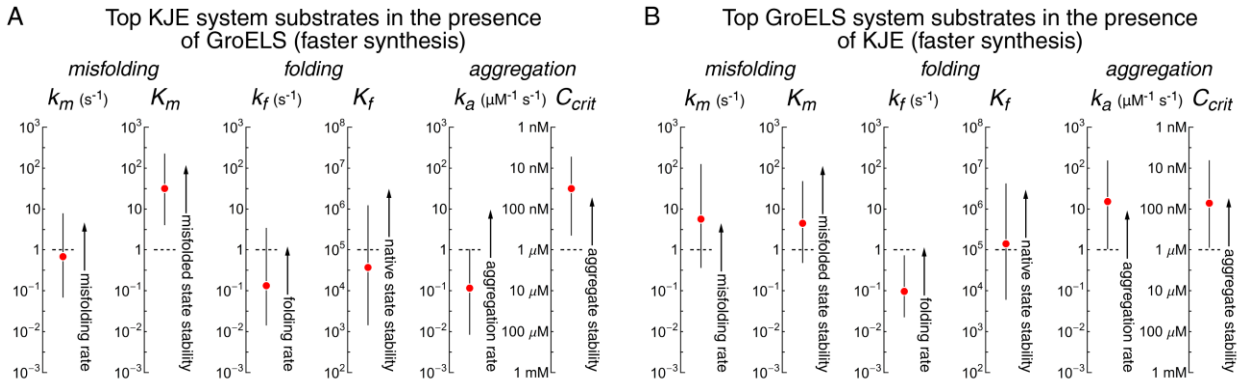


Figure S5, related to Figure 5. Testing FoldEco against Experimental Data on the Effect of Translation Rate on Luciferase Folding. (A) Three-dimensional representation of the tif image of the gel from Figure 2a of Siller et al (Siller et al., 2010). (B) Plot of the fraction of active Luc vs. time during the *in vitro* refolding of Luc (0.25 µM) in the presence of DnaK (1.25 µM), DnaJ (0.25 µM) and GrpE (1.25 µM). Black circles: data from Figure 1 of Szabo et al. (Szabo et al., 1994). Red line: fit of FoldEco to the data (see Extended Experimental Methods); R^2 = 0.98. (C) and (D) As Figure 5A and 5B, respectively, but using the default chaperone concentrations from Table S1 in the FoldEco simulation. (E) Plot of the FoldEco-calculated ratio of the absolute activity of Luc expressed in the slower and faster translating bacteria vs. the fold-increase in k_f in slower translating bacteria (which mimics the effect of cotranslational folding). The experimental value of the former (~1.5) corresponds to a value of 1.9 for the latter.

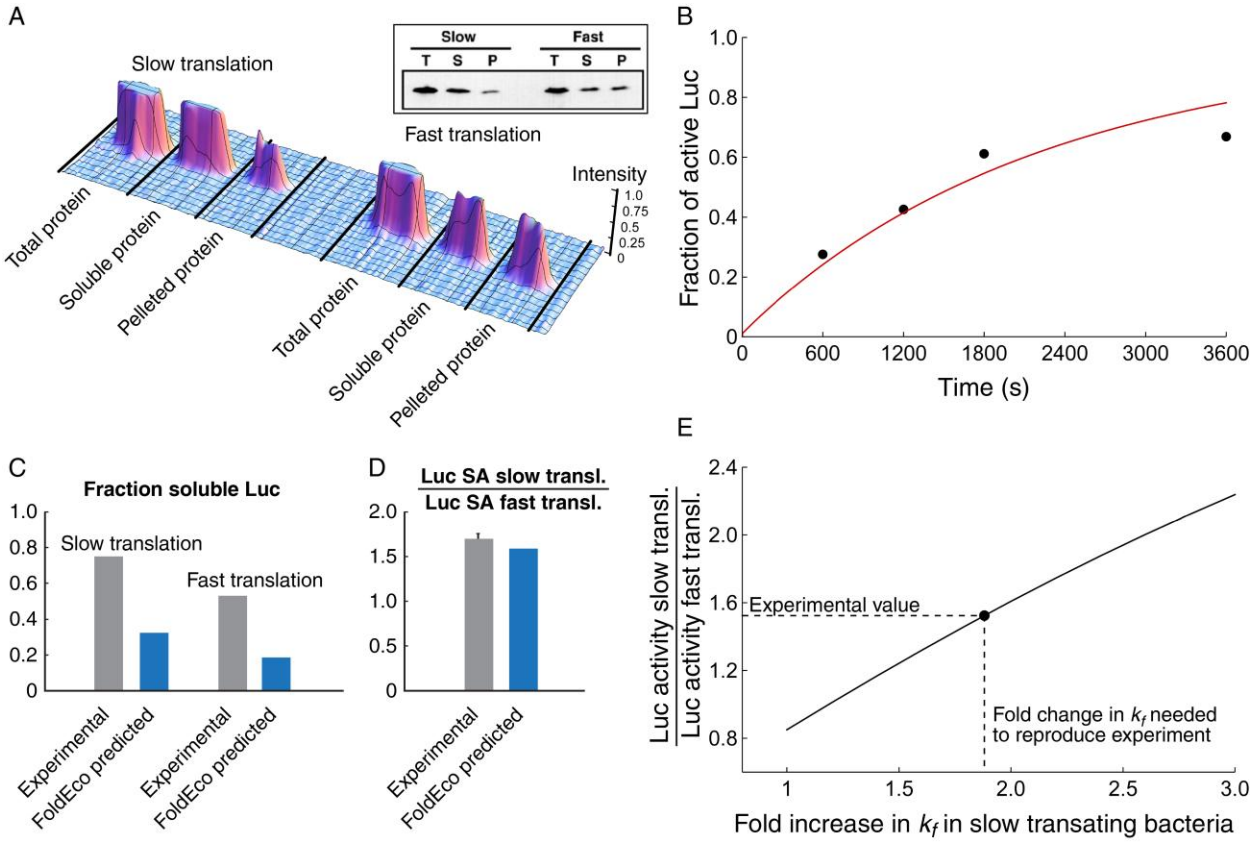


Table Legends (for Tables included as separate Excel files)

Table S3, related to Figure 3. List of biophysical profiles and simulation results for the Monte Carlo simulation experiments with a synthesis rate of 0.02 μM s^{-1}.

Table S5, related to Figure 4. List of biophysical profiles and simulation results for the Monte Carlo simulation experiments with a synthesis rate of 0.2 μM s^{-1}.

Tables

Table S1, related to Figure 1. Concentrations of proteostasis network components. Concentration data for the proteostasis network components in FoldEco were obtained from 5 databases. The primary data are listed for each database. In two cases (Neidhardt and Ellison) this is wt% of total protein. In the other three cases (Marcotte, Frishman, and Xie) this is copy number. For each data source, the primary data are converted to concentrations in μM units. The reported mean is the geometric mean (since concentrations are naturally positive), and the range given is the 50% confidence interval (25th to 75th percentile), assuming that the concentration measurements have a lognormal distribution. The column labeled "FoldEco concs. (μM)" lists the default concentrations used in FoldEco. These are simply round numbers close to the mean values.

	Neidhardt[a]		Ellison[b]		Marcotte[c]		Frishman[d]		Xie[e]		Mean (μM) (50% CI)	FoldEco Concs. (μM)
Component	wt%	Conc (μM)	wt%	Conc (μM)	Copy #	Conc (μM)	Copy #	Conc (μM)	Copy #	Conc (μM)		
Trigger factor	0.61	25.3	0.967	40.1	10043	16.7	31200	51.8	730	4.0	20 (10 to 40)	20
DnaK	0.933	27.0	2.439	70.6	13814	22.9	31200	51.8	2435	13.5	31 (20 to 49)	30
DnaJ	–	–	–	–	1354	2.3	805	1.3	69	0.4	1.0 (0.6 to 1.9)	1
GrpE	0.117	10.7	0.212	19.5	3564	5.9	16000	26.5	–	–	13 (8.6 to 21)	15
GroEL	1.35	47.1	1.09	38.0	11649	19.4	61000	101	–	–	43 (27 to 69)	42
GroES	0.244	47.0	0.32	61.6	64699	107	6540	10.9	2150	11.9	33 (17 to 66)	35
ClpB	–	–	0.298	6.2	341	0.6	1640	2.7	115	0.6	1.6 (0.7 to 3.4)	1.8
Lon	0.21	4.8	0.077	1.8	500	0.8	1550	2.6	159	0.9	1.7 (1.1 to 2.9)	1.8

[a] From Van Bogelen et al (Van Bogelen et al., 1996). Protein weight fractions were measured from *E. coli* K-12 grown in glucose minimal MOPS media at 37 °C with a doubling time of 40 min by quantifying spots in a 2D gel by radiography. The weight fractions were converted to μM concentrations by multiplying them by the concentration of protein in the bacterial cytosol (200 g L^{-1}), and dividing the resulting weight concentration by the relevant molecular weight. The protein concentration in cytoplasm is taken from Cayley, et al (Cayley et al., 1991).

[b] From Lopez-Campistrous, et al (Lopez-Campistrous et al., 2005) These concentrations were measured from *E. coli* K-12 grown "in the presence of amino acids", by differential in gel electrophoresis (DIGE). Weight fractions were converted to μM concentrations as described above.

[c] From Lu et al. (Lu et al., 2007). Protein copy numbers were measured from *E. coli* K-12 grown in glucose minimal MOPS media at 37 °C by "absolute protein expression measurements" (APEX), a mass spectrometric method that relies on counts of peptide fragments from digests of total protein. Copy numbers were converted to μM concentrations by converting them to moles (dividing by Avogadro's number), then dividing them by the volume of cytoplasm in a bacterial cell. Cell volume depends on the rate of cell division; fast-dividing cells are substantially larger than slow-dividing cells. For cells growing in glucose-minimal MOPS media at 37 °C, we used a cell volume of 1 fL and assumed that this was mostly cytoplasm. Cell size was estimated based on the data reported by Kubitschek (Kubitschek, 1969).

[d] From Ishihama, et al. (Ishihama et al., 2008). Protein copy numbers were measured from *E. coli* K-12 MC4100 grown in "rich or minimal media" at 37 °C by "exponentially modified protein abundance index" (emPAI), a mass spectrometric method that relies on counts of peptide fragments from digests of total protein. The data from rich and minimal media were combined. Copy numbers were converted to μM concentrations as described for Marcotte above.

[e] From Taniguchi, et al (Taniguchi et al., 2010). Protein copy numbers were measured from *E. coli* grown in M9 minimal media at 30 °C by single molecule fluorescence measurements. Copy numbers were converted to μM concentrations as described for Marcotte above, except that these authors used exceptionally slow dividing cells in their experiments (they report 4-5 cell divisions in 11-12 hours), and slow dividing bacteria tend to be small. Unfortunately, the authors provide no scale bars in their fluorescence microscopy images, so we cannot calculate the volume directly. They do, however, mention that the surface areas of their bacteria vary from 1.9 to 4.0 μm^2, and assuming that the bacteria are approximately cylindrical with a diameter ~0.6 μm (Woldringh et al., 1977; Zaritsky et al., 1993), this suggests a volume between 0.2 and 0.5 fL. We use 0.3 fL as a compromise.

Table S2, related to Figure 2. Rate constants used in FoldEco simulations for synthesis and degradation rate simulation experiments.

Folding		**KJE system**		**B+KJE system**	
Parameter	*Value*	*Parameter*	*Value*	*Parameter*	*Value*
k_m	1 s^{-1}	$k_{0,1}$	0.1 $\mu M^{-1} s^{-1}$	$b_{i,j,1}$	0.3 $\mu M^{-1} s^{-1}$
K_m	10	$k_{0,2}$	0.0001 s^{-1}	$b_{i,j,2}$	0.006 s^{-1}
k_f	1 s^{-1}	$k_{0,3}$	0.1 $\mu M^{-1} s^{-1}$	$b_{i,j,3}$	1 $\mu M^{-1} s^{-1}$
K_f	1000	$k_{0,4}$	0.2 s^{-1}	$b_{i,j,4}$	2 s^{-1}
k_a	0.1 $\mu M^{-1} s^{-1}$	$k_{i,5}$, $k_{i,17}$	0.3 $\mu M^{-1} s^{-1}$	$b_{i,j,5}$	1 $\mu M^{-1} s^{-1}$
C_{crit}	0.1 μM	$k_{i,6}$, $k_{i,18}$	0.006 s^{-1}	$b_{i,j,6}$	1 $\mu M^{-1} s^{-1}$
$C_{supercrit}$ [a]	10 μM	$k_{i,7}$, $k_{i,19}$	1 $\mu M^{-1} s^{-1}$	$b_{i,j,7}$	0.006 s^{-1}
nucleus size	3	$k_{i,8}$, $k_{i,20}$	2 s^{-1}	$b_{i,j,8}$	2 s^{-1}
Synthesis		$k_{i,9}$, $k_{i,21}$	1 $\mu M^{-1} s^{-1}$	$b_{i,j,9}$	1 s^{-1}
Parameter	*Value*	$k_{i,10}$, $k_{i,22}$	1 $\mu M^{-1} s^{-1}$	$b_{i,j,10}$	1 $\mu M^{-1} s^{-1}$
$s_{0,1}$, $s_{i,5}$	0.02 $\mu M^{-1} s^{-1}$	$k_{i,11}$, $k_{i,23}$	0.006 s^{-1}	$b_{i,j,11}$	0.3 $\mu M^{-1} s^{-1}$
$s_{0,2}$, $s_{i,6}$	0.05 s^{-1}	$k_{i,12}$, $k_{i,24}$	2 s^{-1}	$b_{i,j,12}$	1 s^{-1}
$s_{i,3}$, $s_{i,4}$	0.00050 s^{-1} to 0.375 s^{-1}	$k_{i,13}$	1 s^{-1}	$b_{i,j,13}$	0.1 $\mu M^{-1} s^{-1}$
$s_{i,7}$, $s_{i,8}$	0.25 s^{-1}	$k_{i,14}$	1 $\mu M^{-1} s^{-1}$	$b_{i,j,14}$	0.002 s^{-1}
$s_{i,9}$	0.2 $\mu M^{-1} s^{-1}$	$k_{i,15}$	0.3 $\mu M^{-1} s^{-1}$	$b_{i,j,15}$	0.3 $\mu M^{-1} s^{-1}$
$s_{i,10}$	0.05 s^{-1}	$k_{i,16}$	1 s^{-1}	$b_{i,j,16}$	1 s^{-1}
$s_{i,11}$, $s_{i,12}$	0.1 s^{-1}	**GroELS system**		$b_{i,j,17}$	0.1 s^{-1}
Initial concentrations		*Parameter*	*Value*	**Degradation**	
Species	*Concentration*	$g_{0,1}$	30 s^{-1}	*Parameter*	*Value*
Ribosomes	20 μM	$g_{0,2}$	0.1 s^{-1}	$d_{i,1}$, $d_{i,5}$	0.1 $\mu M^{-1} s^{-1}$
Trigger factor	0 μM	$g_{i,3}$, $g_{i,7}$, $g_{i,k,36} - g_{i,k,41}$	10 $\mu M^{-1} s^{-1}$	$d_{i,2}$, $d_{i,6}$	1 s^{-1}
DnaK	0 μM	$g_{i,4}$, $g_{i,8}$	0.1 s^{-1}	$d_{i,3}$, $d_{i,7}$	5 s^{-1}
DnaJ	0 μM	$g_{i,5}$, $g_{i,9}$	1 $\mu M^{-1} s^{-1}$	$d_{i,4}$	0.1 s^{-1}
GrpE	0 μM	$g_{i,6}$, $g_{i,10}$	0.1 s^{-1}	$d_{i,8}$	0.1 $\mu M^{-1} s^{-1}$
GroEL	0 μM	$g_{i,11}$, $g_{i,12}$, $g_{i,k,42} - g_{i,k,47}$	2.5 s^{-1}	$d_{i,9}$	0.1 s^{-1}
GroES	0 μM	$g_{i,13}$, $g_{i,14}$	1 $\mu M^{-1} s^{-1}$	$d_{i,10}$	0.05 s^{-1}
Lon	0.1 to 20 μM	$g_{i,15}$, $g_{i,22}$, $g_{i,29}$	k_m	$d_{i,11}$	0.1 s^{-1}
Dn (ClpAP-type protease)	0 μM	$g_{i,16}$, $g_{i,23}$, $g_{i,30}$	k_m/K_m		
ClpB	0 μM	$g_{i,17}$, $g_{i,24}$, $g_{i,31}$	k_f		
Client protein (all states)	0 μM	$g_{i,18}$, $g_{i,25}$, $g_{i,32}$	k_f/K_f		
		$g_{i,19}$, $g_{i,20}$, $g_{i,21}$	0.1 s^{-1}		
		$g_{i,26}$, $g_{i,27}$, $g_{i,28}$	30 s^{-1}		
		$g_{i,33}$, $g_{i,34}$, $g_{i,35}$	1 s^{-1}		

[a] $C_{supercrit}$ is the "supercritical concentration", or the monomer dissociation constant from oligomers smaller than the nucleus, as defined by Powers and Powers (Powers and Powers, 2006).

Table S4, related to Figure 4. Rate constants used in FoldEco simulations for Monte Carlo simulation experiments.

Folding		**KJE system**		**B+KJE system**	
Parameter	*Value*	*Parameter*	*Value*	*Parameter*	*Value*
k_m	see Table S3	$k_{0,1}$	0.1 $\mu M^{-1} s^{-1}$	$b_{i,j,1}$	0.3 $\mu M^{-1} s^{-1}$
K_m	see Table S3	$k_{0,2}$	0.0001 s^{-1}	$b_{i,j,2}$	0.006 s^{-1}
k_f	see Table S3	$k_{0,3}$	0.1 $\mu M^{-1} s^{-1}$	$b_{i,j,3}$	1 $\mu M^{-1} s^{-1}$
K_f	see Table S3	$k_{0,4}$	0.2 s^{-1}	$b_{i,j,4}$	2 s^{-1}
k_a	see Table S3	$k_{i,5}$, $k_{i,17}$	0.3 $\mu M^{-1} s^{-1}$	$b_{i,j,5}$	1 $\mu M^{-1} s^{-1}$
C_{crit}	see Table S3	$k_{i,6}$, $k_{i,18}$	0.006 s^{-1}	$b_{i,j,6}$	1 $\mu M^{-1} s^{-1}$
$C_{supercrit}$ [a]	100×C_{crit}	$k_{i,7}$, $k_{i,19}$	1 $\mu M^{-1} s^{-1}$	$b_{i,j,7}$	0.006 s^{-1}
nucleus size	3	$k_{i,8}$, $k_{i,20}$	2 s^{-1}	$b_{i,j,8}$	2 s^{-1}
Synthesis		$k_{i,9}$, $k_{i,21}$	1 $\mu M^{-1} s^{-1}$	$b_{i,j,9}$	1 s^{-1}
Parameter	*Value*	$k_{i,10}$, $k_{i,22}$	1 $\mu M^{-1} s^{-1}$	$b_{i,j,10}$	1 $\mu M^{-1} s^{-1}$
$s_{0,1}$, $s_{i,5}$	0.02 $\mu M^{-1} s^{-1}$	$k_{i,11}$, $k_{i,23}$	0.006 s^{-1}	$b_{i,j,11}$	0.3 $\mu M^{-1} s^{-1}$
$s_{0,2}$, $s_{i,6}$	0.05 s^{-1}	$k_{i,12}$, $k_{i,24}$	2 s^{-1}	$b_{i,j,12}$	1 s^{-1}
$s_{i,3}$, $s_{i,4}$	0.00101 s^{-1} or 0.0116 s^{-1}	$k_{i,13}$	1 s^{-1}	$b_{i,j,13}$	0.1 $\mu M^{-1} s^{-1}$
$s_{i,7}$, $s_{i,8}$	0.25 s^{-1}	$k_{i,14}$	1 $\mu M^{-1} s^{-1}$	$b_{i,j,14}$	0.002 s^{-1}
$s_{i,9}$	0.2 $\mu M^{-1} s^{-1}$	$k_{i,15}$	0.3 $\mu M^{-1} s^{-1}$	$b_{i,j,15}$	0.3 $\mu M^{-1} s^{-1}$
$s_{i,10}$	0.05 s^{-1}	$k_{i,16}$	1 s^{-1}	$b_{i,j,16}$	1 s^{-1}
$s_{i,11}$, $s_{i,12}$	0.1 s^{-1}	**GroELS system**		$b_{i,j,17}$	0.1 s^{-1}
Initial concentrations		*Parameter*	*Value*	**Degradation**	
Species	*Concentration*	$g_{0,1}$	30 s^{-1}	*Parameter*	*Value*
Ribosomes	20 μM	$g_{0,2}$	0.1 s^{-1}	$d_{i,1}$, $d_{i,5}$	0.1 $\mu M^{-1} s^{-1}$
Trigger factor	20 μM	$g_{i,3}$, $g_{i,7}$, $g_{i,k,36} - g_{i,k,41}$	10 $\mu M^{-1} s^{-1}$	$d_{i,2}$, $d_{i,6}$	1 s^{-1}
DnaK	0 or 30 μM	$g_{i,4}$, $g_{i,8}$	0.1 s^{-1}	$d_{i,3}$, $d_{i,7}$	5 s^{-1}
DnaJ	0 or 1 μM	$g_{i,5}$, $g_{i,9}$	1 $\mu M^{-1} s^{-1}$	$d_{i,4}$	0.1 s^{-1}
GrpE	0 or 15 μM	$g_{i,6}$, $g_{i,10}$	0.1 s^{-1}	$d_{i,8}$	0.1 $\mu M^{-1} s^{-1}$
GroEL	0 or 42 μM	$g_{i,11}$, $g_{i,12}$, $g_{i,k,42} - g_{i,k,47}$	2.5 s^{-1}	$d_{i,9}$	0.1 s^{-1}
GroES	0 or 35 μM	$g_{i,13}$, $g_{i,14}$	1 $\mu M^{-1} s^{-1}$	$d_{i,10}$	0.05 s^{-1}
Lon	0.3 μM	$g_{i,15}$, $g_{i,22}$, $g_{i,29}$	k_m	$d_{i,11}$	0.1 s^{-1}
Dn (ClpAP-type protease)	0 μM	$g_{i,16}$, $g_{i,23}$, $g_{i,30}$	k_m/K_m		
ClpB	0 μM	$g_{i,17}$, $g_{i,24}$, $g_{i,31}$	k_f		
Client protein (all states)	0 μM	$g_{i,18}$, $g_{i,25}$, $g_{i,32}$	k_f/K_f		
		$g_{i,19}$, $g_{i,20}$, $g_{i,21}$	0.1 s^{-1}		
		$g_{i,26}$, $g_{i,27}$, $g_{i,28}$	30 s^{-1}		
		$g_{i,33}$, $g_{i,34}$, $g_{i,35}$	1 s^{-1}		

[a] $C_{supercrit}$ was set to be 100-fold higher than C_{crit}; it was not varied independently.

Table S6, related to Figure 5. Results of FoldEco simulations of firefly luceiferase expression experiments by Siller et al. at the 15 min time point.

Slower Translation Rate							
Protein state	**Total (µM)**	**Free (µM)**	**TF bound (µM)**	**KJE bound[a] (µM)**	**GroELS bound (µM)**	**B+KJE bound[a] (µM)**	**Lon bound[b] (µM)**
U	38.73	2.23	15.89	20.10	0.50	N/A	0
M	20.39	0.99	6.94	8.00	4.46	N/A	0
N	24.86	24.80	N/A	N/A	0.06	N/A	
A	52.37	2.30	N/A	N/A	N/A	50.07	
Total	136.35	30.33	22.83	5.02	5.02	50.07	0
Faster Translation Rate							
Protein state	**Total (µM)**	**Free (µM)**	**TF bound (µM)**	**KJE bound[a] (µM)**	**GroELS bound (µM)**	**B+KJE bound[a] (µM)**	**Lon bound[b] (µM)**
U	36.98	2.09	16.17	18.22	0.50	N/A	0
M	19.63	0.93	7.00	7.24	4.46	N/A	0
N	33.94	33.89	N/A	N/A	0.06	N/A	
A	171.24	7.27	N/A	N/A	N/A	163.96	
Total	261.79	44.18	23.17	25.47	5.01	163.96	0

[a]The column ”KJE bound” includes only unfolded and misfolded protein bound to DnaK and $DnaJ_2$. Aggregates bound to DnaK and $DnaJ_2$ are accounted for in the column “B+KJE bound”, along with aggregates bound to ClpB. For aggregates, we give the size-weighted concentration of aggregates (i.e., the concentration of monomers in the aggregated state).

[b]Includes only protein that is reversibly bound to Lon. Protein that is already committed to degradation is not included.

Table S7, related to Figure 5. Results of FoldEco Simulations with Lower Chaperone Concentrations of Firefly Luceiferase Expression Experiments by Siller et al. at the 15 min Time Point.

Slower Translation Rate							
Protein state	**Total (μM)**	**Free (μM)**	**TF bound (μM)**	**KJE bound[a] (μM)**	**GroELS bound (μM)**	**B+KJE bound[a] (μM)**	**Lon bound[b] (μM)**
U	14.54	1.22	4.66	8.35	0.31	N/A	0
M	10.23	0.71	2.59	4.31	2.62	N/A	0
N	20.60	20.57	N/A	N/A	0.03	N/A	
A	95.11	4.95	N/A	N/A	N/A	90.17	
Total	140.49	27.45	7.26	12.66	2.95	90.17	0
Faster Translation Rate							
Protein state	**Total (μM)**	**Free (μM)**	**TF bound (μM)**	**KJE bound[a] (μM)**	**GroELS bound (μM)**	**B+KJE bound[a] (μM)**	**Lon bound[b] (μM)**
U	14.92	1.33	5.53	7.74	0.31	N/A	0
M	9.71	0.71	2.73	3.66	2.61	N/A	0
N	24.58	24.55	N/A	N/A	0.03	N/A	
A	217.10	12.45	N/A	N/A	N/A	204.65	
Total	266.31	39.04	8.25	11.40	2.96	204.65	0

[a]The column "KJE bound" includes only unfolded and misfolded protein bound to DnaK and $DnaJ_2$. Aggregates bound to DnaK and $DnaJ_2$ are accounted for in the column "B+KJE bound", along with aggregates bound to ClpB. For aggregates, we give the size-weighted concentration of aggregates (i.e., the concentration of monomers in the aggregated state).

[b]Includes only protein that is reversibly bound to Lon. Protein that is already committed to degradation is not included.

Supplemental Experimental Procedures

Assumptions and Simplifications in FoldEco

Models like FoldEco that are intended to capture the complexity of biological processes occurring *in vivo* inevitably contain some assumptions and simplifications; without them, it is sometimes impossible to make progress. Here we present a list of the most important assumptions and simplifications in the current version of FoldEco.

FoldEco is a deterministic, not a stochastic, model

FoldEco consists of a set of ordinary differential equations that are numerically integrated to produce a solution that consists of the time-dependent concentrations of the species in the model. As such, FoldEco is valid for examining the average state of proteostasis in populations of *E. coli* under a given set of conditions. Focusing on population averages means that FoldEco cannot capture cell-to-cell variability, but it has the advantage that measuring population averages for variables of interest is often experimentally easier than measuring the same variables in individual cells.

FoldEco assumes free mixing of components

FoldEco treats *E. coli* as single compartments (the cytoplasm) that are freely mixed; that is, we assume that the concentrations of the species in the model do not vary across the cell. Many factors could cause concentration heterogeneities within cells. However, the data to parameterize the spatial dependence of *in vivo* protein concentrations is only now becoming available (Beck et al., 2011), and some of it suggests that FoldEco components like GroEL (Charbon et al., 2011) and DnaK and DnaJ (Taniguchi et al., 2010) are in fact uniformly distributed. Thus, rather than complicating our model by adding the capacity to account for spatially heterogeneous concentrations of FoldEco components as well as introducing parameters for which we do not have estimates, we use the free mixing approximation.

Parameters measured *in vitro* are used for processes occurring *in vivo*

The default parameters used in FoldEco were all measured *in vitro*. However, FoldEco is meant to simulate these processes *in vivo*. Some features of the *in vivo* environment are easy to mimic *in vitro* (pH, ionic strength), but some are not; in particular, the presence of the cellular proteome is expected to affect protein folding and protein association energetics (Ebbinghaus and Gruebele, 2011; Gershenson and Gierasch, 2011). Measurements of protein stabilities *in vivo* have suggested that they are often similar to *in vitro* stabilities (Ghaemmaghami and Oas, 2001; Ignatova and Gierasch, 2004), but they are sometimes higher (Dhar et al., 2011; Ebbinghaus et al., 2010), and high protein concentrations can also lower protein stability (Miklos et al., 2011). Given this uncertainty, we have taken a conservative approach and chosen

to use the *in-vitro*-measured parameters without correction. These parameters will be replaced in the model as specific information about rate or equilibrium constants for the processes in FoldEco becomes available.

FoldEco does not account for bacterial growth

In its present form, FoldEco does not account for bacterial growth, which in effect dilutes proteins as they are being synthesized. Thus, FoldEco best represents bacteria that are not growing. This condition is met by bacteria that are overexpressing heterologous proteins, for which growth is strongly inhibited (Dong et al., 1995; Kurland and Dong, 1996), and bacteria in the lag and stationary phases. Future versions of FoldEco will account for bacterial growth.

FoldEco does not account for the presence of the background proteome

The FoldEco simulations reported herein track the synthesis and folding of a single client protein. However, no protein is produced in isolation: there is a background proteome that in principle could compete for proteostasis network components. This problem is acute during bacterial growth, as many, if not most, newly synthesized proteins interact with the proteostasis network (Calloni et al., 2012; Chapman et al., 2006; Fujiwara et al., 2010; Kerner et al., 2005) and would therefore compete with a client protein of interest. The problem is much reduced when bacteria are not growing, as the proteostasis network does not interact to nearly the same extent with pre-existing proteins (Calloni et al., 2012). Thus, again, FoldEco is most appropriate for use in situations like the overexpression of heterologous proteins, or the lag and stationary phases, where *E. coli* are producing few proteins other than the one of interest. In ongoing work, we are extending FoldEco to account for the proteome, using a realistic parameterization and coarse graining methods.

FoldEco does not have the proteostasis network components for membrane or periplasmic proteins

Membrane proteins have specialized systems for their folding and quality control (Dalbey et al., 2011; Knowles et al., 2009), as do periplasmic proteins (Allen et al., 2009). These systems are not included in the current version of FoldEco, which is intended for use with cytoplasmic proteins, but will be included in future versions.

FoldEco does not account for stress responses

FoldEco does not yet account for stress responses, like the heat shock response (Guisbert et al., 2008), which can upregulate the levels of proteostasis network components in response to environmental changes that affect protein folding. However, such stress responses can be roughly approximated by increasing the initial concentrations of the relevant components and accounting for the effect of the stress on the protein folding parameters (for example,

increasing the temperature would be expected to destabilize a protein). Future versions of FoldEco will incorporate stress responses.

FoldEco does not account for fluctuations in ATP concentration

In this version of FoldEco, we have assumed that the concentration of ATP is in the millimolar range and is constant, so that ATP binding is never rate limiting. However, FoldEco can be used to track ATP use by the proteostasis network, since ATP consumption rates are known for the systems in FoldEco (Figure S1).

Concentrations of mRNA are assumed to be at steady state in FoldEco

The rate constant for translation initiation, $s_{i,3}$, is assumed to be constant throughout FoldEco simulations. This is equivalent to assuming that the concentration of mRNA for the client protein of interest is at steady state at the beginning of the simulation. This assumption is consistent with observations that transcription is not rate limiting in protein production (Sanden et al., 2003) that protein synthesis lags mRNA synthesis (Golding et al., 2005), and that protein translation can occur co-transcriptionally (Miller et al., 1970). However, given this assumption, one must be judicious in choosing the t = 0 time point when simulating experimental data. The t = 0 time point should be assigned to the time at which the client protein mRNA (and therefore the rate of protein synthesis) reaches steady state, not necessarily the time when expression is induced. When mRNA levels reach steady state rapidly compared to the time scale of the simulation (say, within 15 min for a 3 hour simulation), the time when expression is induced can be assigned to be t = 0. This condition should be satisfied most of the time, given that the approach of the concentration of a given mRNA to its steady state value is determined by its degradation rate, and the half-lives of mRNA in *E. coli* are generally 3-8 min (Bernstein et al., 2002). When this condition is not met, the time point at which the protein synthesis rate reaches steady state should be assigned to be t = 0.

FoldEco uses a simplified model of translation

In FoldEco, protein synthesis is represented as a simplified, three-step process. The synthesis of a client protein (protein “i”) begins with the initiation of translation. This process involves several steps (Laursen et al., 2005), but is simplified to a single step in FoldEco. This simplification is appropriate since there is a single rate limiting step in translation initiation: the rearrangement to the stable 30S initiation complex of the metastable ternary complex between the 30S ribosome subunit, mRNA, and the formyl-Met-charged tRNA (Gualerzi et al., 1977; Laursen et al., 2005). The next step is mRNA translation up to the point that the nascent protein is long enough to emerge from the ribosomal exit tunnel, and the final step is translation of the remainder of the mRNA. We treat both of these processes as single steps, rather than treating the translation of individual codons separately, as has been done in some models of translation

(Zouridis and Hatzimanikatis, 2007). This approach is justified here, because the rate of protein synthesis rapidly comes to steady state (within ~1 min for all of the simulations described in this work) on the time scales of interest to us (15 min to 2.75 h). Using a finer grained model of translation is therefore not necessary.

FoldEco does not explicitly account for co-translational folding and misfolding

Many proteins are known to begin folding co-translationally, especially multi-domain proteins (Eichmann et al., 2010; Evans et al., 2008; Fedorov and Baldwin, 1995; Frydman et al., 1999; Kelkar et al., 2012; Siller et al., 2010). The role of co-translational folding in E. coli, however, is diminished by two factors. First, bacterial ribosomes translate proteins much faster than eukaryotic ribosomes, leaving less time for co-translational folding (Siller et al., 2010). Second, trigger factor and/or DnaK bind to most proteins as they emerge from the ribosomal exit tunnel (Deuerling et al., 1999; Teter et al., 1999; Wegrzyn and Deuerling, 2005). This role of trigger factor and DnaK is incorporated implicitly into the model by not including a pathway for co-translational folding or misfolding. This approach should yield acceptable results, except in the case of double deletions of trigger factor and DnaK. Future versions of FoldEco will explicitly include the possibility of co-translational folding and misfolding.

Aggregation in FoldEco

Aggregates composed of two, three, etc. monomers and all of their complexes with chaperones (of which there are nine) count as individual species in FoldEco. The size of the system of equations in FoldEco therefore depends on the maximum aggregate size allowed. We have found that artifacts arise when the cut-off size is too small, while computation times are too long when the cut-off size is too large (computation time scales quadratically with the number of equations being solved: with a maximum aggregate size of 100, FoldEco can be solved in ~1 s, but with a maximum size of 1000, 20-40 s are needed). We find that a maximum aggregate size between 50 and 200 is a good compromise between these considerations. An aggregate consisting of 200 monomers is much smaller than typical inclusion bodies (Taylor et al., 1986), but increasing the size cut-off does not change the behavior of the model substantially (see below for more information on the interpretation of the maximum fibril size setting).

The stoichiometry of B+KJE system components binding to aggregates is assumed to be 1:1

In principle, it should be possible for multiple ClpB molecules to be working in parallel on a single aggregate (provided that the aggregate is large enough). However, in the mechanism for the B+KJE disaggregation system (Figure S1D), a 1:1 stoichiometry is assumed between disaggregation system components and aggregates. Unfortunately, this is a computational necessity. Each aggregate size is tracked individually in the model; thus, if aggregates composed of up to n monomers can each have m ClpB molecules bound to them, then $n \times m$ equations

would be needed to track all of the aggregate:ClpB species, which would greatly increase computation times. This problem is mitigated, however, upon consideration of how FoldEco represents aggregation. The most intuitive way to view the aggregated states A_2, A_3,…, A_n is as individual oligomers. In a rough approximation, they also could be viewed as reactive sites on a single large aggregate instead, in which case the number concentration of aggregates ( ) is equivalent to the number of reactive sites on the single aggregate, the size-weighted concentration of aggregates ( ) is equivalent to the total number of monomers in the single aggregate, and *n* corresponds to the maximum number of monomers in the aggregate per reactive site. In this scenario, ClpB molecules can attach to each reactive site, and the density of reactive sites is controlled by adjusting *n*, with the number of reactive sites decreasing as *n* increases.

Proteostasis network components not currently in FoldEco

Some chaperoning systems are not yet included in FoldEco, like the bacterial small heat shock proteins, IbpA and IbpB, and the bacterial Hsp90, HtpG. However, the effects of these systems on proteostasis appear to be subtle (Bardwell and Craig, 1988; Thomas and Baneyx, 1998). Nevertheless, future versions of FoldEco will incorporate these proteostasis network components as the necessary mechanistic information becomes available.

Synthesis and Degradation Rate Simulation Experiments

We determined how the balance between synthesis and degradation affects aggregation in the absence of chaperones in FoldEco by seeking conditions that would allow a model client protein with k_f = 1 s^{-1}, K_f = 1000, k_m = 1 s^{-1}, K_m = 10, k_a = 0.1 μM^{-1} s^{-1}, and C_{crit} = 0.1 μM (the other parameters used are listed Table S2) to be less than 5% aggregated after 10,000 s of simulation time. To do this, the chaperone concentrations were set to 0 and FoldEco was solved for the model protein at a constant synthesis rate while the Lon concentration was varied (using a quasi-Newton method as implemented by the Mathematica® function FindMinimum) until the amount of aggregates at 10,000 s was 5% of total protein. Above this Lon concentration, in the unshaded region of Figure 2A, aggregates are < 5% of total protein; below it, in the shaded region of Figure 2A, aggregates are > 5%. This was repeated at several synthesis rates, ranging from 0.01 μM s^{-1} to 1 μM s^{-1}, to give the points that, when joined, give the black curve in Figure 2A. FoldEco was solved again with the synthesis rates and Lon concentrations found above to determine the native protein concentration and the fraction of synthesized protein that has been degraded at 10,000 s. The synthesis rate was varied by varying $s_{i,3}$ and $s_{i,4}$ between 0.000504 s^{-1} and 0.375 s^{-1} with $s_{i,7}$ = $s_{i,8}$ = 0.25 s^{-1}, $s_{i,11}$ = $s_{i,12}$ = 0.1 s^{-1} (appropriate for synthesis of a 200-residue protein at 20 residues s^{-1}; Figure S1A), and a ribosome concentration of 20 μM. This procedure was repeated with the default chaperone concentrations (Table S1) to

determine how including chaperones modulated the effect of the balance between synthesis and degradation on aggregation model client protein described above.

Approach to steady state

We solved FoldEco for the test protein described in the "Synthesis and Degradation Rate Simulation Experiments", which has the following set of folding parameters: k_f = 1 s^{-1}, K_f = 1000, k_m = 1 s^{-1}, K_m = 10, k_a = 0.1 μM^{-1} s^{-1}, and C_{crit} = 0.1 μM, and an initial Lon concentration of 0.3 μM. The other parameters used are listed Table S2. The solutions for unfolded, misfolded and native protein are shown in Figure S2B. These plots show that protein accumulation occurs on two time scales. It begins rapidly, with the unfolded and misfolded states reaching a pseudo-steady state within 50 s. The concentrations of these species slowly increase after that until they reach their true steady state at 15,000-20,000 s.

Analysis of a simplified model for protein synthesis, folding, and degradation

We analyzed the behavior of FoldEco on the two time scales described above using a simplified model that includes only synthesis, folding, and degradation (Figure S2C). Synthesis is condensed to a single step with rate σ. Degradation is simplified to a Briggs-Haldane scheme, where D is a Lon-type protease that degrades U and M. The rate constants for substrate binding, release, and degradation (δ_1, δ_2, δ_3) are assumed to be the same for U and M. The folding and unfolding rate constants are φ_1 and φ_2, while those for misfolding and un-misfolding are μ_1 and μ_2. The rate equations for the system are

—— (S1)

—— (S2)

—— (S3)

——— (S4)

——— (S5)

—— (S6)

There is also one conservation equation:

(S7)

We begin by examining the true steady state, where all of the rate equations above are equal to 0. Setting equations S4 and S5 equal to 0 and rearranging them gives

(S8)

(S9)

where the subscript "s" indicates steady state. Summing equations (S1) and (S2), and equations (S4) and (S5) gives

(S10)

(S11)

And from equation (S3) we have

(S12)

Summing equations (S10), (S11), and (S12) gives

Or

(S13)

Inserting this into equation (S8) for the conservation of D and rearranging gives

(S14)

So, the concentration of free protease can be written simply in terms of the total protease concentration ($[D]_{tot}$), the synthesis rate (σ), and the equivalent of k_{cat} for the protease (δ_3). In addition, combining equations (S10), (S12), (S13), and (S14) and rearranging gives

(S15)

For convenience we define

(S16)

Δ_s is the effective degradation rate constant. Equation (S15) can now be written as

$$\text{—} \qquad \text{(S17)}$$

We note that if $\sigma/\delta_3 > [D]_{tot}$, then $[D]_s$ and Δ_s would be negative, which is not physically reasonable. In this situation, synthesis and degradation cannot be balanced, and there is no steady state. The amount of protein in the system simply increases without bound.

Equation (S17) shows that the sum of the concentrations of U and M at true steady state can be written in terms of the synthesis and degradation parameters, independent of the folding parameters. The ratio of $[U]_s$ and $[M]_s$ can be obtained by inserting equation (S9) into equation (S2) to give

$$\text{——} \qquad \text{————} \qquad \text{(S18)}$$

Or

$$\text{———————} \qquad \text{(S19)}$$

The ratio of [M] and [U] at steady state is then

$$\text{——} \quad \text{——} \qquad \text{(S20)}$$

At equilibrium, in the absence of synthesis and degradation, this ratio would be

$$\text{——} \quad \text{—}$$

Since Δ_s has to be positive if steady state is possible, the steady state ratio of [M] and [U] has to be lower than their equilibrium ratio. In other words, the degradation system decreases the apparent steady-state stability of the misfolded state. Finally, the steady state ratio of [N] and [U] is

$$\text{——} \quad \text{—} \qquad \text{(S21)}$$

The steady state ratio of [N] and [U] is the same as it is at equilibrium.

We analyze the earlier pseudo-steady state by assuming that the rate equations for all of the species in the system except N are approximately equal to 0, and that $\varphi_1[U] >> \varphi_2[N]$ (i.e., unfolding is negligible compared to folding). The equations for processes that do not involve

interconversion or interaction with the native state retain their form, so that equations (S8) and (S9) become

(S22)

(S23)

where the subscript "ps" indicates pseudo-steady state. We begin our analysis of the pseudo-steady state by summing equations (S1) and (S4), and equations (S2) and (S5) to get

(S24)

(S25)

Inserting the expressions for [D:U] and [D:M] into equations (S24) and (S25) gives

(S26)

(S27)

The coefficients in the last terms in these equations are similar to the expression for Δ_s. We therefore define

(S28)

where $[D]_{ps}$ is the pseudo-steady state concentration of D. Equations (S26) and (S27) can be rearranged and written in matrix form

(S29)

The determinant of the rate constant matrix $\boldsymbol{R_{ps}}$ is

(S30)

The solution to equation (S29) is

(S31)

From this solution, the sum of [U] and [M] at pseudo-steady state is

(S32)

This is similar in form to the expression at steady state in equation (S17), except that there is a second term in the denominator, and it depends explicitly on folding parameters: the folding rate φ_1 and the quantity $\mu_1/(\mu_2 + \Delta_{ps})$. This latter quantity is the pseudo-steady state ratio of [M] to [U], as is evident from equation (S31):

(S33)

Again, Δ_{ps} has to be positive, so the ratio of [M] and [U] at pseudo-steady state is less than it is at equilibrium, just as it is at true steady state. Thus, degradation decreases the apparent steady-state stability of the misfolded state throughout the time course of protein synthesis, and not just at steady state.

The effect of this apparent destabilization of the misfolded state on the concentration of the native state is mediated by its effect on the concentration of the unfolded state. The concentration of native protein at steady state is proportional to the concentration of unfolded protein, as illustrated by rearranging equation (S21).

(S34)

The rate of accumulation of native protein at pseudo-steady state is also proportional to the concentration of unfolded protein. Given the pseudo-steady state assumptions, the rate equation for [N] becomes

(S35)

Because degradation destabilizes the misfolded state relative to the unfolded state at both pseudo- and true steady state, the concentration of unfolded protein is higher than one would expect based on its equilibrium energetics under these conditions. As a consequence, both the rate of accumulation of natively folded protein and its final concentration are higher because of degradation. This result explains the observation that jointly increasing the synthesis rate and protease concentration can result in an increase in the native state concentration throughout the time course of a FoldEco simulation.

As noted above and as apparent from Figure S2B, protein accumulation in FoldEco happens in two stages. The pseudo-steady state is reached rapidly, and is followed by a much slower evolution to true steady state. The rate of approach to true steady state cannot be determined

analytically, even for this simplified system. We can, however, get some information by examining the asymptotic rate of approach to steady state using perturbation methods. To do this, we first observe that the equations for [D:U] and [D:M] have the same form at pseudo- and true steady state, and so it is reasonable to assume that they retain this form throughout the approach to steady state. Thus,

(S36)

(S37)

where the subscript "t" indicates that the equations are valid between pseudo- and true steady state. Combining equations (S36) and (S37) with equation (S7) yields

(S38)

Inserting equations (S36) and (S37) into equations (S1), (S2), and (S3) gives

(S39)

(S40)

(S41)

Close to steady state, $[U]_t$ differs from $[U]_s$ by only a small quantity u:

(S42)

Similarly, for $[M]_t$, $[D]_t$, and $[N]_t$:

(S43)

(S44)

(S45)

The value of d can be approximated in terms of u and m using equation (S38) and noting that $[D]_t = [D]_s$ at steady state

(S46)

Using equations (S14) and (S15) to substitute for $[U]_s + [M]_s$ enables us to simplify equation (S46) to

(S47)

We note that the last two terms in equation (S39) can be combined to give

(S48)

Using equations (S42) and (S43),

(S49)

Neglecting the last term in which the small quantities *u* and *d* are multiplied, and using equation (S47) to replace *d* gives

(S50)

Inserting equation (S16) into the above gives

(S51)

Similarly,

(S52)

With this groundwork, we can write equations for *du*/*dt*, *dm*/*dt*, and *dn*/*dt* for the asymptotic rate of approach of the system to steady state. Inserting equations (S42)-(S45) into equations (S39)-(S41) and then simplifying with equations (S51) and (S52) gives the following:

(S53)

(S54)

(S55)

In matrix form,

(S56)

Let the matrix of rate constants above be denoted $\boldsymbol{R_p}$, where the subscript "p" indicates that it is for the rate equations of the perturbations u, m, and n. This system of equations is first-order, homogeneous, linear, and has constant coefficients. Its solutions are therefore sums of exponentials, the coefficients and exponents of which are derived from the eigenvectors and eigenvalues of $\boldsymbol{R_p}$. The eigenvalue of $\boldsymbol{R_p}$ with the smallest magnitude is of particular interest because it determines the rate of approach of the system to steady state. Unfortunately, the eigenvalues of a 3×3 matrix are the roots of a third-order polynomial, and are therefore difficult to analyze. A lower bound on these eigenvalues, however, can be determined using the Gershgorin circle theorem (Marcus and Minc, 1964). For an n×n matrix $\boldsymbol{A}$ with elements a_{ij}, we can define

(S57)

The Gershgorin circle theorem states that each eigenvalue of $\boldsymbol{A}$ is in at least one of n circles in the complex plane centered at a_{ii} and with radius r_i:

(S58)

For $\boldsymbol{R_p}$, a 3×3 matrix, there are 3 circles. Since the diagonal elements of $\boldsymbol{R_p}$ are real, these circles are centered on the real number axis. Furthermore, $\boldsymbol{R_p}$ is diagonally similar to a symmetric matrix, so its eigenvalues are real as well. The bounds on the eigenvalues are therefore the points where the Gershgorin circles intersect the real number axis. To shrink the circles we can use a diagonal similarity transformation, wherein $\boldsymbol{R_p}$ is multiplied by a diagonal matrix $\boldsymbol{P}$ and its inverse $\boldsymbol{P}^{-1}$ on the right and left, respectively.

(S59)

This operation leaves the eigenvalues of $\boldsymbol{R_p}$ unchanged, but changes the radii of the Gershgorin circles, so that the bounds can be improved. If we take

(S60)

then

(S61)

and

(S62)

where c is a positive constant. The value of c can be chosen so that all of the Gershgorin circles lie entirely in the left half of the complex plane. With this choice, we can now concern ourselves exclusively with the lowest-magnitude (least negative) bound on the eigenvalues, that is, the intersection of the Gershgorin circles with the real number axis that is closest to 0. The intersections of the Gershgorin circle with the real number axis are given by the center ± the radius of the circles. Since the circles are entirely in the left-half of the complex plane, the intersection for which the center and radius are added is closest to 0. The sum of the centers and radii of the Gershgorin circles happens to be equal to the column sums of the matrix. These column sums are

(S63)

(S64)

(S65)

Here we have used equations (S14) and (S17) to simplify these column sums. Since c can be chosen arbitrarily, we can use it to ensure that the column 1 and column 3 sums are equal:

(S66)

Rearranging equation (S66) shows that

(S67)

The left-hand side of equation (S67) is equal to φ_1 when $c = 0$ and increases to $\varphi_1 + \varphi_2$ when $c = 1$. The right-hand side is equal to 0 when $c = 0$ and tends to infinity when $c \rightarrow 1-$. Thus, the left- and right-hand sides of equation (S67) have to be equal somewhere in the range $0 \leq c \leq 1$. With this information, we can rearrange equation (S66) as follows:

(S68)

In equation (S68), the second term has to be negative, because it was shown above that $0 \leq c \leq 1$. This observation means that

(S69)

In this inequality, the left-hand side is the bound on the eigenvalue from column 2 of $\boldsymbol{R_p}$. The right-hand side is the bound on the eigenvalue from column 3 of $\boldsymbol{R_p}$ (which is equal to the bound from column 1 by virtue of our choice of c). Inequality (S69) shows that the lowest magnitude bound on the eigenvalues of $\boldsymbol{R_p}$, which we will denote β_{min}, is therefore the one derived from columns 1 and 3, that is:

(S70)

The rate of approach of the system to steady state clearly depends on φ_2, the rate constant for unfolding of the native state. It also depends on the value of c. An explicit expression for c can be obtained by rearranging equation (S66):

(S71)

This equation is quadratic in c. The positive root is:

(S72)

This root can be simplified by noting that $\varphi_1 >> \varphi_2$ for proteins with stable native states (recall that the equilibrium constant for folding, K_f, is equal to φ_1/φ_2 and is usually >> 1). Since $0 \leq c \leq 1$, the term $\varphi_2 c^2$ in equation (S71) can be neglected, and the root simplifies to

(S73)

Inserting this into equation (S70) gives

(S74)

Equation (S74) shows that when $\Delta_s[D]_s/[D]_{tot} >> \varphi_1$, β_{min} is

(S75)

which happens when degradation is much faster than folding. On the other hand, when $\varphi_1 >> \Delta_s[D]_s/[D]_{tot}$,

(S76)

which happens when folding is much faster than degradation. In summary, the rate of approach of the system to steady state is governed by the smallest eigenvalue of $\boldsymbol{R_p}$, which is the apparent rate constant for the approach of the system to steady state. The bound on this eigenvalue is β_{min}, which is a function of the folding, unfolding, and degradation rates, but is never larger than the unfolding rate constant of the client protein. In FoldEco, the unfolding rate constant is given by k_f / K_f, the ratio of the folding rate and equilibrium constants.

Examination of KJE and GroELS substrates

Four thousand random biophysical profiles were generated, comprising folding rate and equilibrium constants (k_f and K_f), misfolding rate and equilibrium constants (k_m and K_m), monomer-aggregate association rate constants (k_a), and critical concentrations for aggregation (C_{crit}). The values of k_f and k_m varied between 10^{-3} and 10^3 s^{-1}; k_a varied between 10^{-3} and 10^3 μM^{-1} s^{-1}; K_f varied between 10^2 and 10^8; K_m varied between 10^{-3} and 10^3; and C_{crit} varied between 1 mM and 1 nM. In all cases, the parameters were log-uniformly distributed (i.e., their logarithms had a uniform distribution) between the indicated bounds.

The bounds for k_f are taken from a set of reported folding rates of single domain proteins (Ouyang and Liang, 2008; Plaxco et al., 1998). The same bounds are used for k_m, since misfolding and folding should be kinematically similar, despite having different endpoints. The bounds for k_a correspond to nonspecific bimolecular associations at the low end to the diffusion limit at the high end (Northrup and Erickson, 1992). Values of K_f from 10^2 to 10^8 correspond to folding free energies from about −3 to −11 kcal mol^{-1}, which are typical of single domain proteins (although the very low end of stabilities would probably only be found in proteins with destabilizing mutations) (Ghosh and Dill, 2010; Ghosh and Dill, 2009). Values of K_m from 10^{-3} to 10^3 correspond to misfolding free energies from about +4 to −4 kcal mol^{-1}. This range reflects our expectation that the misfolded state should be unstable for well-behaved proteins, but usually less stable than the native state even for poorly-behaved proteins. The range for C_{crit} (which is also equal to the monomer-aggregate dissociation constant) corresponds to weak, nonspecific protein-protein associations at the low end to moderately strong protein-protein associations at the high end (Kastritis et al., 2011).

While using the bounds described above ensures that individual parameters will have realistic values, it is possible that combinations of parameters that do not occur in nature will be

included in our data set. Such parameter combinations, however, can still be instructive about trends in chaperone behavior. Moreover, we are unaware of experimental data on joint parameter distributions that could be used to justify blocking out regions of parameter space. We therefore uniformly sampled the entire six-dimensional parameter space.

FoldEco was solved at two synthesis rates (0.02 μM s^{-1} and 0.2 μM s^{-1}) with each of the random biophysical profiles under three conditions: with neither the KJE nor the GroELS system, with the KJE system (30 μM DnaK, 1 μM DnaJ, 15 μM GrpE), or with the GroELS system (3 μM GroEL tetradecamer, 5 μM GroES hexamer). The desired synthesis rates were obtained by setting $s_{i,3}$ and $s_{i,4}$ to 0.001014 s^{-1} or 0.011628 s^{-1}, with $s_{i,7} = s_{i,8} = 0.25$ s^{-1}, $s_{i,11} = s_{i,12} = 0.1$ s^{-1} (appropriate for synthesis of a 200-residue protein at 20 residues s^{-1}; Figure S1A). In all cases, Lon (0.3 μM hexamer) and trigger factor (20 μM) were included, but ClpB and native state proteases were excluded. The same association rate and equilibrium constants with DnaK, DnaJ, and GroEL (those given in Figure S1) were used for all of the random client proteins, and the folding and misfolding rates were assumed to be the same inside the GroEL cavity as in solution. We provide a complete list of the parameters used in Table S4 (other than k_f, K_f, k_m, K_m, k_a, and C_{crit} which are in Tables S3 and S5). Of the 4,000 solutions at the low synthesis rate, 13 (0.325%) failed error tests (i.e., the solutions had insufficient accuracy or precision) at time steps before 10,000 s. At the high synthesis rate, 11 (0.275%) solutions failed error tests. These problems arise almost exclusively in the rare cases in which a very fast aggregation rate ($k_a > 10^{2.5}$ μM^{-1} s^{-1}) is coupled with low fibril stability (C_{crit} > 100 μM), and tendencies toward high misfolded state stability (K_m > 1) and a low folding rate (k_f < 1 s^{-1}). For this small region of parameter space, the solutions to FoldEco are too stiff to integrate up to 10,000 s with the default accuracy and precision goals, even with Mathematica's powerful numerical differential equation solving algorithms. This problem could be addressed by relaxing the accuracy and/or precision goals, but since such a small portion of the parameter space was affected, we simply discarded the cases in question. The solutions were evaluated at 10,000 s and the top 5% of the increases in the concentration of the native state upon adding the KJE system or the GroELS system (top 200) were identified (Tables S3 and S5).

Testing FoldEco against Experimental Data on the Effect of Translation Rate on Luciferase Folding

Siller et al. examined the effect of translation rate on the *in vivo* folding efficiency of firefly luciferase (Luc), an aggregation-prone protein (Siller et al., 2010). They controlled the translation rate using streptomycin pseudo-dependent (Sm^P) ribosomes of *E. coli*, which have a translation rate of 5 amino acids per second (AA s^{-1}) under normal conditions, but an approximately two-fold higher translation rate in the presence of streptomycin (Ruusala et al., 1984; Siller et al., 2010; Zengel et al., 1977). They found that the aggregation of Luc decreased

at the slower translation rate (in the absence of streptomycin), while the specific and absolute activities of Luc increased, after both 15 min and 1 h of Luc expression. Our goal in this experiment was to determine to what extent these experimental results can be reproduced in simulations with FoldEco.

To do this, we first needed quantitative information on all of these effects. The extents of aggregation in equivalent amounts of Luc produced at slower or faster translation rates (here, "slower" and "faster" translation rates refers to the translation rates of Sm^P ribosomes in the absence or presence of streptomycin, or about 5 or 10 AA s^{-1}, respectively) after 15 min or 1 h of expression are illustrated by a western blot (15 min; their Figure 2a) or a Coomassie-stained gel (1 h; their Figure 2b). These gels qualitatively show that there is less aggregation with slower translation of Luc than with faster translation. To quantify these results, we copied the images of the gels in the pdf file of the Siller et al. paper using Adobe® Acrobat®, pasted the image into Adobe® Photoshop®, and saved them as grayscale tif files. These files, which each consisted of a set of numbers between 0 and 1 that represent the gray level of the image (0 = black, 1 = white), were imported into Mathematica® and inverted (so that 1 = black and 0 = white). The result for the gel from the 15 min time point is shown as a three-dimensional plot in Figure S5A, with the original gel shown as an inset. The bands representing Luc in the lanes with soluble and pelleted protein were integrated using Mathematica® to determine the relative amounts of Luc in each of these bands. The resulting integrated intensities (in arbitrary units) for the Luc bands in the gel from the 15 min time point after background subtraction were: soluble Luc, slow translation = 188.9; pelleted Luc, slow translation = 64.2; soluble Luc, fast translation = 96.4; pelleted luc, fast translation = 86.6. Thus, the fraction of Luc that remained soluble at the slow translation rate (f_{sol}) was 0.75. This fraction decreased to 0.53 at the faster translation rate. Unfortunately, the Luc bands were not sufficiently resolved from background bands in the gel from the 1 h time point to reliably integrate them, so from here on we focused on the data from the 15 min time point.

The relative specific activity (r_{SA}) of Luc at 15 min was determined directly from the bar graph in Figure 2a in Siller et al (Siller et al., 2010). This graph shows the relative Luc activity in a given amount of total protein and the Luc activity present in just the soluble fraction when Luc is expressed at slow or fast translation rates. We focused on the relative activities of the soluble fractions, since they had virtually all of the Luc activity. Measuring directly from the bar graph in Figure 2a in Siller et al., we found that the ratio of the relative activities in the soluble fractions from samples of Luc expressed at slower and faster translation rates was r_{SA} = 1.7.

To simulate these experimental results using FoldEco, we first had to constrain the input parameters to the extent possible. It has been shown that Luc folds *in vitro* via a mechanism that involves an initial fast equilibration between the unfolded state and a pair of aggregation-

prone intermediates, which we consider together to make up the misfolded state (M) in FoldEco. This first event happens within the dead time of manual mixing experiments. Based on this observation, we assigned k_m = 1 s^{-1} (Herbst et al., 1998) The stability of these two intermediates collectively was measured to be ~4 kcal mol^{-1}, which corresponds to K_m = 800 (Herbst et al., 1997). The overall stability of Luc was measured to be ~12 kcal mol^{-1}, which corresponds to $K_f = 5 \times 10^8$ (Herbst et al., 1997). Luc aggregates at concentrations below 100 nM, so we assigned C_{crit} = 10 nM; however, this process is not fast at sub-micromolar concentrations, so we assigned k_a = 0.1 µM s^{-1} (Herbst et al., 1998; Herbst et al., 1997). The aggregation of Luc shows some signs of behaving like a downhill polymerization—the concentration of soluble aggregation-prone intermediate in equilibrium with aggregates has a weak tendency to increase in concentration as a function of total protein (Herbst et al., 1998)—so the supercritical concentration was set equal to C_{crit}.

To constrain k_f, we took advantage of literature data for the *in vitro* KJE-system-mediated refolding of Luc. Szabo et al. measured the kinetics of reactivation of Luc (0.25 µM) in the presence of 1.25 µM DnaK, 0.25 µM DnaJ, and 1.25 µM GrpE (Szabo et al., 1994). Their data (fraction of active Luc vs. time) were read off of their Figure 1 and are reproduced in Figure S5B. FoldEco was then used to fit this data set by setting the concentrations of the KJE components to their experimental values, and setting the concentrations of the ribosome, proteases, and the other chaperones equal to 0. With these settings, all of the systems in FoldEco are switched off except the KJE cycle, and FoldEco simplifies to a model of KJE-system-assisted *in vitro* folding. The folding parameters were set to the values noted above for Luc, and the parameters in Table S2 were used for the steps in the KJE cycle. FoldEco was then fit to the experimental data using the Mathematica® function NonlinearModelFit with k_f as the only adjustable parameter. The fit to the data was good (R^2 = 0.98; Figure S5B), with the best-fit value of k_f being 0.015 s^{-1} (95% confidence interval: 0.010 s^{-1} to 0.023 s^{-1}). This value of k_f completed the biophysical profile of Luc,

The parameters for protein synthesis were constrained as follows. Siller et al. reported that the rate of protein synthesis in their system was proportional to the rate of polypeptide elongation (steps $s_{i,7}$ and $s_{i,8}$, and $s_{i,11}$ and $s_{i,12}$ in Figure S1A). Protein synthesis was therefore rate limited by this process rather than translation initiation. The rate of elongation was set to 5 AA s^{-1} for the slower translating bacteria and 10 AA s^{-1} for the faster translating bacteria (Ruusala et al., 1984). According to Figure S1A, this makes $s_{i,7} = s_{i,8}$ = 0.125 s^{-1} and $s_{i,11} = s_{i,12}$ = 0.0098 s^{-1} for the slower translating bacteria, and $s_{i,7} = s_{i,8}$ = 0.25 s^{-1} and $s_{i,11} = s_{i,12}$ = 0.0196 s^{-1} for the faster translating bacteria. The rate constant for the initiation of translation ($s_{i,3}$) was set to 0.1 s^{-1}, so that initiation was much faster than elongation. Figure 1A in Siller et al suggests that their bacteria double in ~1 h during log-phase growth for the faster translating bacteria (Siller et al., 2010), so the concentration of ribosomes (which correlates with the growth rate) for these

bacteria was set to 20 μM (Bremer and Dennis, 1987). The concentration of ribosomes was kept the same for the slower translating bacteria, since their slow growth rate was assumed to be due to their translation rate, not the concentration of their ribosomes.

The concentrations of the other proteostasis network components were set using the information in Table S1. This table shows that reported measurements of chaperone vary widely, and any set of concentrations within the 50% confidence intervals in Table S1 can be supported with experimental data. Because the expression of aggregation-prone proteins like Luc is known to induce the heat-shock response (Hoffmann and Rinas, 2004), we used concentrations from the tops of these confidence intervals, which are comparable to the concentrations reported by Ishihama et al. and Lopez-Campistrous et al. (Ishihama et al., 2008; Lopez-Campistrous et al., 2005). Thus we set the chaperone concentrations as follows: [trigger factor] = 40 μM, [DnaK] = 50 μM, [DnaJ] = 2 μM, [GrpE] = 20 μM, [GroEL tetradecamer] = 5 μM, [GroES heptamer] = 10 μM, [Lon hexamer] = 0.5 μM, [ClpB hexamer] = 0.5 μM. Siller et al. found no difference in chaperone concentrations between the slower and faster translating bacteria (Siller et al., 2010), so we used the above chaperone concentrations in our simulations of both conditions.

Using the parameters noted above and the default values in Table S2 for the remaining parameters, we simulated the expression of Luc in the slower and faster translating bacteria. We took the fraction of soluble Luc (f_{sol}) to be

A substantial amount of non-native chaperone-bound Luc was included in the soluble fraction. Siller et al. do not report experiments that can be used to judge whether their soluble protein in fact included chaperone-bound, non-native Luc (Siller et al., 2010). The ratio of specific activities of Luc in the slower and faster translating bacteria (r_{SA}) was taken to be

The results of our simulation at the 15 min time point are summarized in Table S6, according to which the values of f_{sol} for the slower and faster translating bacteria are 0.62 and 0.35, and the value of r_{SA} is 1.4. These values are compared to the experimental results in Figure 5.

To determine the effect of chaperone concentrations on our results, the simulations were repeated with the default chaperone concentrations from Table S1 ([trigger factor] = 20 μM, [DnaK] = 30 μM, [DnaJ] = 1 μM, [GrpE] = 15 μM, [GroEL tetradecamer] = 3 μM, [GroES

heptamer] = 5 μM, [Lon hexamer] = 0.3 μM, [ClpB hexamer] = 0.3 μM). The results of these simulations at lower chaperone concentrations are summarized in Table S7, according to which the values of f_{sol} for the slower and faster translating bacteria are 0.32 and 0.18, and the value of r_{SA} is 1.6. These values are compared to the experimental results in Figure S5C and D. As stated in the main text, the simulations reproduce the trends in the experimental data, but they overemphasize aggregation.

Of the proteostasis network components in FoldEco, the results of the Luc expression simulations are most sensitive to the concentrations of the KJE system components (data not shown). The concentration of GroEL does not strongly affect our results, consistent with the experimentally observed inability of GroEL to facilitate Luc refolding *in vitro* (Buchberger et al., 1996).

Siller et al. observe that not only the specific activity, but the absolute activity of Luc increases in the slower translating bacteria. According to their Figure 2c, the Luc produced by the slower translating bacteria is about 1.5-fold higher than that produced by the faster translating bacteria after 1 h (Siller et al., 2010). They attribute this effect to improved cotranslational folding when translation is slower. Since FoldEco does not explicitly account for cotranslational folding (see above), this effect is not reproduced in our simulations. We can, however, roughly approximate cotranslational folding by increasing the k_f in the simulation with the slower translation rate. We find that increasing k_f by a factor of about 1.9 reproduces the experimentally observed 1.5-fold difference in the absolute activity of Luc produced in the slower translating bacteria (Figure S5E). Thus, co-translational folding increases the effective folding rate of Luc by 1.9 fold.

Supplemental References

Allen, W.J., Phan, G., and Waksman, G. (2009). Structural biology of periplasmic chaperones. Adv Protein Chem Struct Biol *78*, 51-97.

Bardwell, J.C., and Craig, E.A. (1988). Ancient heat shock gene is dispensable. J. Bacteriol. *170*, 2977-2983.

Beck, M., Topf, M., Frazier, Z., Tjong, H., Xu, M., Zhang, S., and Alber, F. (2011). Exploring the spatial and temporal organization of a cell's proteome. J Struct Biol *173*, 483-496.

Bernstein, J.A., Khodursky, A.B., Lin, P.H., Lin-Chao, S., and Cohen, S.N. (2002). Global analysis of mRNA decay and abundance in Escherichia coli at single-gene resolution using two-color fluorescent DNA microarrays. Proc Natl Acad Sci U S A *99*, 9697-9702.

Bremer, H., and Dennis, P.P. (1987). Modulation of chemical composition and other parameters of the cell by growth rate. In *Escherichia coli* and *Salmonella typhimurium*: Cellular and molecular biology, F.C. Neidhardt, ed. (Washington, D. C., American Society for Microbiology), pp. 1527-1542.

Buchberger, A., Schroder, H., Hesterkamp, T., Schonfeld, H.J., and Bukau, B. (1996). Substrate shuttling between the DnaK and GroEL systems indicates a chaperone network promoting protein folding. J Mol Biol *261*, 328-333.

Calloni, G., Chen, T., Schermann, S.M., Change, H.-C., Genevaux, P., Agostini, F., Tartaglia, G.G., Hayer-Hartl, M., and Hartl, F.U. (2012). DnaK functions as a central hub in the *E. coli* chaperone network. Cell Reports, In press.

Cayley, S., Lewis, B.A., Guttman, H.J., and Record, M.T., Jr. (1991). Characterization of the cytoplasm of *Escherichia coli* K-12 as a function of external osmolarity. Implications for protein-DNA interactions in vivo. J. Mol. Biol. *222*, 281-300.

Chapman, E., Farr, G.W., Usaite, R., Furtak, K., Fenton, W.A., Chaudhuri, T.K., Hondorp, E.R., Matthews, R.G., Wolf, S.G., Yates, J.R*., et al.* (2006). Global aggregation of newly translated proteins in an Escherichia coli strain deficient of the chaperonin GroEL. Proc Natl Acad Sci U S A *103*, 15800-15805.

Charbon, G., Wang, J., Brustad, E., Schultz, P.G., Horwich, A.L., Jacobs-Wagner, C., and Chapman, E. (2011). Localization of GroEL determined by in vivo incorporation of a fluorescent amino acid. Bioorg Med Chem Lett *21*, 6067-6070.

Dalbey, R.E., Wang, P., and Kuhn, A. (2011). Assembly of bacterial inner membrane proteins. Annu Rev Biochem *80*, 161-187.

Deuerling, E., Schulze-Specking, A., Tomoyasu, T., Mogk, A., and Bukau, B. (1999). Trigger factor and DnaK cooperate in folding of newly synthesized proteins. Nature *400*, 693-696.

Dhar, A., Girdhar, K., Singh, D., Gelman, H., Ebbinghaus, S., and Gruebele, M. (2011). Protein stability and folding kinetics in the nucleus and endoplasmic reticulum of eucaryotic cells. Biophys J *101*, 421-430.

Dong, H., Nilsson, L., and Kurland, C.G. (1995). Gratuitous overexpression of genes in Escherichia coli leads to growth inhibition and ribosome destruction. J Bacteriol *177*, 1497-1504.

Ebbinghaus, S., Dhar, A., McDonald, J.D., and Gruebele, M. (2010). Protein folding stability and dynamics imaged in a living cell. Nat Methods *7*, 319-323.

Ebbinghaus, S., and Gruebele, M. (2011). Protein Folding Landscapes in the Living Cell. J Phys Chem Lett *2*, 314-319.
Eichmann, C., Preissler, S., Riek, R., and Deuerling, E. (2010). Co-translational structure acquisition of nascent polypeptides monitored by NMR spectroscopy. Proc Natl Acad Sci U S A *107*, 9111-9116.
Evans, M.S., Sander, I.M., and Clark, P.L. (2008). Co-translational folding promotes beta-helix formation and avoids aggregation in vivo. J Mol Biol *383*, 683-692.
Fedorov, A.N., and Baldwin, T.O. (1995). Contribution of co-translational folding to the rate of formation of native protein structure. Proc Natl Acad Sci U S A *92*, 1227-1231.
Frydman, J., Erdjument-Bromage, H., Tempst, P., and Hartl, F.U. (1999). Co-translational domain folding as the structural basis for the rapid de novo folding of firefly luciferase. Nat Struct Biol *6*, 697-705.
Fujiwara, K., Ishihama, Y., Nakahigashi, K., Soga, T., and Taguchi, H. (2010). A systematic survey of *in vivo* obligate chaperonin-dependent substrates. EMBO J. *29*, 1552-1564.
Gershenson, A., and Gierasch, L.M. (2011). Protein folding in the cell: challenges and progress. Curr Opin Struct Biol *21*, 32-41.
Ghaemmaghami, S., and Oas, T.G. (2001). Quantitative protein stability measurement in vivo. Nat Struct Biol *8*, 879-882.
Ghosh, K., and Dill, K. (2010). Cellular proteomes have broad distributions of protein stability. Biophys. J. *99*, 3996-4002.
Ghosh, K., and Dill, K.A. (2009). Computing protein stabilities from their chain lengths. Proc. Natl. Acad. Sci. U. S. A. *106*, 10649-10654.
Golding, I., Paulsson, J., Zawilski, S.M., and Cox, E.C. (2005). Real-time kinetics of gene activity in individual bacteria. Cell *123*, 1025-1036.
Gualerzi, C., Risuleo, G., and Pon, C.L. (1977). Initial rate kinetic analysis of the mechanism of initiation complex formation and the role of initiation factor IF-3. Biochemistry *16*, 1684-1689.
Guisbert, E., Yura, T., Rhodius, V.A., and Gross, C.A. (2008). Convergence of molecular, modeling, and systems approaches for an understanding of the Escherichia coli heat shock response. Microbiol Mol Biol Rev *72*, 545-554.
Herbst, R., Gast, K., and Seckler, R. (1998). Folding of firefly (*Photinus pyralis*) luciferase: aggregation and reactivation of unfolding intermediates. Biochemistry *37*, 6586-6597.
Herbst, R., Schafer, U., and Seckler, R. (1997). Equilibrium intermediates in the reversible unfolding of firefly (*Photinus pyralis*) luciferase. J Biol Chem *272*, 7099-7105.
Hoffmann, F., and Rinas, U. (2004). Stress induced by recombinant protein production in *Escherichia coli*. Adv Biochem Eng Biotechnol *89*, 73-92.
Ignatova, Z., and Gierasch, L.M. (2004). Monitoring protein stability and aggregation in vivo by real-time fluorescent labeling. Proc Natl Acad Sci U S A *101*, 523-528.
Ishihama, Y., Schmidt, T., Rappsilber, J., Mann, M., Hartl, F.U., Kerner, M.J., and Frishman, D. (2008). Protein abundance profiling of the *Escherichia coli* cytosol. BMC Genomics *9*, 102.
Kastritis, P.L., Moal, I.H., Hwang, H., Weng, Z., Bates, P.A., Bonvin, A.M., and Janin, J. (2011). A structure-based benchmark for protein-protein binding affinity. Protein Sci. *20*, 482-491.
Kelkar, D.A., Khushoo, A., Yang, Z., and Skach, W.R. (2012). Kinetic Analysis of Ribosome-bound Fluorescent Proteins Reveals an Early, Stable, Co-translational Folding Intermediate. J Biol Chem *287*, 2568-2578.

Kerner, M.J., Naylor, D.J., Ishihama, Y., Maier, T., Chang, H.C., Stines, A.P., Georgopoulos, C., Frishman, D., Hayer-Hartl, M., Mann, M.*, et al.* (2005). Proteome-wide analysis of chaperonin-dependent protein folding in *Escherichia coli*. Cell *122*, 209-220.
Knowles, T.J., Scott-Tucker, A., Overduin, M., and Henderson, I.R. (2009). Membrane protein architects: the role of the BAM complex in outer membrane protein assembly. Nat Rev Microbiol *7*, 206-214.
Kubitschek, H.E. (1969). Growth during the bacterial cell cycle: analysis of cell size distribution. Biophys. J. *9*, 792-809.
Kurland, C.G., and Dong, H. (1996). Bacterial growth inhibition by overproduction of protein. Mol Microbiol *21*, 1-4.
Laursen, B.S., Sorensen, H.P., Mortensen, K.K., and Sperling-Petersen, H.U. (2005). Initiation of protein synthesis in bacteria. Microbiol Mol Biol Rev *69*, 101-123.
Lopez-Campistrous, A., Semchuk, P., Burke, L., Palmer-Stone, T., Brokx, S.J., Broderick, G., Bottorff, D., Bolch, S., Weiner, J.H., and Ellison, M.J. (2005). Localization, annotation, and comparison of the *Escherichia coli* K-12 proteome under two states of growth. Mol. Cell. Proteomics *4*, 1205-1209.
Lu, P., Vogel, C., Wang, R., Yao, X., and Marcotte, E.M. (2007). Absolute protein expression profiling estimates the relative contributions of transcriptional and translational regulation. Nat. Biotechnol. *25*, 117-124.
Marcus, M., and Minc, H. (1964). A survey of matrix theory and matrix inequalities (New York, Dover Publications).
Miklos, A.C., Sarkar, M., Wang, Y., and Pielak, G.J. (2011). Protein crowding tunes protein stability. J Am Chem Soc *133*, 7116-7120.
Miller, O.L., Jr., Hamkalo, B.A., and Thomas, C.A., Jr. (1970). Visualization of bacterial genes in action. Science *169*, 392-395.
Northrup, S.H., and Erickson, H.P. (1992). Kinetics of protein-protein association explained by Brownian dynamics computer simulation. Proc. Natl. Acad. Sci. U. S. A. *89*, 3338-3342.
Ouyang, Z., and Liang, J. (2008). Predicting protein folding rates from geometric contact and amino acid sequence. Protein Sci. *17*, 1256-1263.
Plaxco, K.W., Simons, K.T., and Baker, D. (1998). Contact order, transition state placement and the refolding rates of single domain proteins. J. Mol. Biol. *277*, 985-994.
Powers, E.T., and Powers, D.L. (2006). The kinetics of nucleated polymerizations at high concentrations: amyloid fibril formation near and above the "supercritical concentration". Biophys. J. *91*, 122-132.
Ruusala, T., Andersson, D., Ehrenberg, M., and Kurland, C.G. (1984). Hyper-accurate ribosomes inhibit growth. EMBO J *3*, 2575-2580.
Sanden, A.M., Prytz, I., Tubulekas, I., Forberg, C., Le, H., Hektor, A., Neubauer, P., Pragai, Z., Harwood, C., Ward, A.*, et al.* (2003). Limiting factors in Escherichia coli fed-batch production of recombinant proteins. Biotechnol Bioeng *81*, 158-166.
Siller, E., DeZwaan, D.C., Anderson, J.F., Freeman, B.C., and Barral, J.M. (2010). Slowing bacterial translation speed enhances eukaryotic protein folding efficiency. J. Mol. Biol. *396*, 1310-1318.
Szabo, A., Langer, T., Schroder, H., Flanagan, J., Bukau, B., and Hartl, F.U. (1994). The ATP hydrolysis-dependent reaction cycle of the Escherichia coli Hsp70 system DnaK, DnaJ, and GrpE. Proc Natl Acad Sci U S A *91*, 10345-10349.

Taniguchi, Y., Choi, P.J., Li, G.W., Chen, H., Babu, M., Hearn, J., Emili, A., and Xie, X.S. (2010). Quantifying *E. coli* proteome and transcriptome with single-molecule sensitivity in single cells. Science *329*, 533-538.
Taylor, G., Hoare, M., Gray, D.R., and Marston, F.A.O. (1986). Size and density of protein inclusion bodies. Bio-Technol *4*, 553-557.
Teter, S.A., Houry, W.A., Ang, D., Tradler, T., Rockabrand, D., Fischer, G., Blum, P., Georgopoulos, C., and Hartl, F.U. (1999). Polypeptide flux through bacterial Hsp70: DnaK cooperates with trigger factor in chaperoning nascent chains. Cell *97*, 755-765.
Thomas, J.G., and Baneyx, F. (1998). Roles of the *Escherichia coli* small heat shock proteins IbpA and IbpB in thermal stress management: comparison with ClpA, ClpB, and HtpG *in vivo*. J. Bacteriol. *180*, 5165-5172.
Van Bogelen, R.A., Abshire, K.Z., Pertsemlidis, A., Clark, R.L., and Neidhardt, F.C. (1996). Gene-protein database of *Escherichia coli* K-12, edition 6. In Escherichia coli and Salmonella: cellular and molecular biology, F.C. Neidhardt, R. Curtiss III, J.L. Ingraham, E.C.C. Lin, K.B. Low, B. Magasanik, W.S. Reznikoff, M. Riley, M. Schaechter, and H.E. Umbarger, eds. (Washington, D. C., ASM Press), pp. 2067-2117.
Wegrzyn, R.D., and Deuerling, E. (2005). Molecular guardians for newborn proteins: ribosome-associated chaperones and their role in protein folding. Cell Mol Life Sci *62*, 2727-2738.
Woldringh, C.L., de Jong, M.A., van den Berg, W., and Koppes, L. (1977). Morphological analysis of the division cycle of two *Escherichia coli* substrains during slow growth. J. Bacteriol. *131*, 270-279.
Zaritsky, A., Woldringh, C.L., Helmstetter, C.E., and Grover, N.B. (1993). Dimensional rearrangement of *Escherichia coli* B/r cells during a nutritional shift-down. J. Gen. Microbiol. *139*, 2711-2714.
Zengel, J.M., Young, R., Dennis, P.P., and Nomura, M. (1977). Role of ribosomal protein S12 in peptide chain elongation: analysis of pleiotropic, streptomycin-resistant mutants of Escherichia coli. J Bacteriol *129*, 1320-1329.
Zouridis, H., and Hatzimanikatis, V. (2007). A model for protein translation: polysome self-organization leads to maximum protein synthesis rates. Biophys. J. *92*, 717-730.